The planar matching endpoints and their leading constant

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Abstract

For two independent uniform samples of size N in the unit square, the minimum over permutations of the maximum of the horizontal quadratic Luxemburg norm and the vertical supremum has leading constant $1/\sqrt{2\pi}$ at scale $\sqrt{\log N/N}(\log\log\log N)^{1/4}$, in probability. The same holds with the axes exchanged. We prove matching Gaussian divergence bounds, using a state-dependent local-time clock for the lower bound and a moving phase with a sharp covariance estimate for the upper bound. We prove the local estimates for both constructions and transfer the Gaussian constant to the point matching problem.

1 Statement

Let $X_1,\dots,X_N$ and $Y_1,\dots,Y_N$ be independent uniform samples in $\Omega=[0,1]^2$. For a permutation π, write $\Delta_i(j)=X_j^{(i)}-Y_{\pi(j)}^{(i)}$. Use the Young function $\Theta(t)=e^{t^2}-1$ and the Luxemburg convention that the modular at the chosen scale is at most one, with mass $1/N$ per pair. Put

$$M_N=\min_{\pi}\max\{\|\Delta_1\|_{\psi_2},\|\Delta_2\|_{\infty}\}.$$

Theorem 1. *With natural logarithms,*

$$\frac{M_N}{\sqrt{\log N/N}\,(\log\log\log N)^{1/4}}\xrightarrow{\mathbb{P}}\frac{1}{\sqrt{2\pi}}.$$

The same conclusion holds for $\min_\pi\max\{\|\Delta_1\|_{\infty},\|\Delta_2\|_{\psi_2}\}$.

The Gaussian result uses continuum currents with an exact zero-normal divergence condition. For centered white noise W^0 on Ω, write P_hW^0 for its dyadic histogram. We use the flux gauge

$$\mathcal{F}(f)=\inf_{\operatorname{div}J=f,\ J\cdot n=0}\max\{\|J_1\|_{\psi_2(dx)},\|J_2\|_{L^\infty(dx)}\}.$$

The infimum is over integrable fields on the square with weak divergence f and zero outer normal flux. Thus $\int f\varphi=-\int J\cdot\nabla\varphi$ for every smooth test on the closed square. Both norms use Lebesgue measure on the unit square. By white noise we mean the centered Gaussian linear process $W(g)$, $g\in L^2(\Omega)$, with covariance $\mathbb{E}[W(g)W(k)]=\int gk$. Its spatial centering is $W^0(g)=W(g-\int g)$. A histogram P_hW^0 is the density constant on the dyadic squares of side h with those white noise masses.

Theorem 2. *As $h \downarrow 0$ through dyadic widths,*

$$\frac{\mathcal{F}(P_h W^0)}{\sqrt{\log(1/h)}\,(\log\log\log(1/h))^{1/4}} \xrightarrow{\mathbb{P}} \frac{1}{\sqrt{2\pi}}.$$

The expected gauge has the same asymptotic.

The normalization has two independent factors. A planar Poisson block has coordinate covariance $1/(4\pi)$ per unit of natural log-scale. The determinant-one control has optimal balanced coefficient $\sqrt{2}$ in the Brownian variance clock. The two samples supply a $\sqrt{2}$ noise factor and their spatial cutoff has $\log(1/h) \sim \frac{1}{2}\log N$, which cancel in the final normalization.

Here and below, all asymptotics are as the indicated mesh tends to zero or the sample size tends to infinity. Constants in $O(\cdot)$ and $\lesssim$ may depend on parameters explicitly fixed before this limit.

The lower construction tests the divergence against a potential with positive vertical derivative. Size bias by this derivative makes the ratio of the two derivatives a martingale. The clock for that martingale determines the lower coefficient. For the upper construction, a determinant-one chart changes the covariance of each Gaussian current. A convex phase determines the chart and controls both coordinate costs. We develop the local constructions and repair estimates before using them. The last section derives the Gaussian comparison at the required precision and proves Theorem 1.

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2 Predictable charts and local divergence inverses

A chart on a physical cell has the form $z \mapsto Az + b$, where $\det A = 1$. Write $G = AA^{\mathsf{T}}$ for its metric. A source f and current J in chart coordinates push forward to $f(A^{-1}(x-b))$ and $AJ(A^{-1}(x-b))$, respectively. The change of variables in the weak divergence identity proves that this push-forward preserves divergence and zero normal flux. If $K_0^{-1}G \preceq G_+ \preceq K_0 G$, the two chart changes have operator norm at most $\sqrt{K_0}$. Indeed the squared singular values of $A_+^{-1}A$ are the eigenvalues of $A_+^{-1}GA_+^{-\mathsf{T}}$, which lie in $[K_0^{-1}, K_0]$.

Lemma 1 (Predictable Gaussian bridges)**.** *In a finite nested sequence of partitions, choose each new partition measurably from the earlier Gaussian cell masses. Conditional on the current history, the unrevealed noise in each current cell is an independent white noise bridge on that cell. In particular, once a refinement is chosen, its new contrast has covariance equal to the orthogonal projection from child constants onto the subspace of zero parent means. Two successive refinements chosen together can reveal their contrasts independently, conditional on the old history.*

Proof. A bridge on Q is isonormal on $\{g \in L^2(Q) : \int_Q g = 0\}$, extended by zero on constants. The statement starts with this decomposition on the initial partition, or on the whole square if only the global mean is known. Conditional on a fixed old history, the child indicators are fixed functions in this Hilbert space after subtracting their parent averages. Its orthogonal decomposition into the span of these centered indicators and its orthogonal complement gives two independent Gaussian families. The complement is the orthogonal direct sum of the mean-zero L^2 spaces of the children. This supplies the next conditional law and proves the claim by induction. This argument also constructs the conditional laws for random partitions. For each realized old history use the indicated Gaussian projection kernels on a countable determining set of L^2 tests, and then use mean-square completion for additional tests. The geometric partitions below have measurable indicator maps into L^2 and measurable projections, so these kernels are measurable. A second predetermined refinement adds an orthogonal subspace to the first and hence gives independent contrasts. An area-preserving chart is an isometry on the test spaces. □

Lemma 2 (Whitney subdivision with a minimum side)**.** *Let Q be a parallelogram of diameter D containing a ball of radius $c_0 D$. Fix a minimum side $s > 0$ and a cap θD with $0 < \theta < 1$. For a sufficiently large constant C_0 depending only on c_0, if $\theta D < C_0 s$ we may stop without splitting. Otherwise there is a deterministic measurable subdivision into square children and terminal children. Assign to a square its side ℓ_S, and to a terminal child the value $\ell_S = s$. Then*

$$|S| \asymp \ell_S^2, \qquad \operatorname{diam}(S) \lesssim \ell_S, \qquad \ell_S \asymp \max\{s, \min(\theta D, d_Q(x))\}, \quad x \in S, \qquad d_Q(x) = \operatorname{dist}(x, \partial Q).$$

The constants are uniform in θ. For a sufficiently small fixed $c > 0$, each neighborhood $S^+ = \{x \in Q : \operatorname{dist}(x, S) < c\ell_S\}$ meets only a bounded number of children, of comparable sides. The neighborhoods have bounded overlap. If $l < l'$ are two sides occurring in Q, the children of side l lie within distance Cl of ∂Q. Thus a ball of radius Cl' can meet at most $C'l'/l$ such children, also after enlarging their diameters by a fixed factor.

Proof. Use the fixed dyadic lattice with smallest side s, and choose maximal lattice squares contained in Q, of side $2s \le \ell \le \theta D$, and of boundary clearance at least $\gamma\ell$. Here γ is a large fixed constant. They are disjoint up to null sets. The remaining intersections with lattice s-squares cover the remaining area. To handle a clipped s-square, take a point x in its intersection with Q and move a fraction As/D of the way to the inball center. Convexity guarantees an inball of radius $Ac_0 s$ at the new point. Choose A fixed so this radius exceeds $3s$, and C_0 large enough so the fraction is less

than one. The s-square containing this new point is full and within $O(s)$ of the clipped one. It is also within $O(s)$ of ∂Q. Choose γ larger than this last constant so the full square cannot belong to a selected square. Assign each clipped piece to such an uncovered full square, with ties broken in lattice order. Lattice packing bounds the number of assignments per receiver. Every uncovered full square has boundary distance $O(s)$, because its $2s$-ancestor is under the cap and fails containment or clearance. This proves the terminal estimates, including the area lower bound. The construction is measurable since square containment, clearance and receiver comparisons are measurable, and the bounded lattice search and tie rules can be fixed in advance.

For a selected square, its dyadic parent either fails containment or clearance, giving $d_Q \le C\ell_S$ throughout the child, or exceeds the cap, giving $\ell_S > \theta D/2$. Clearance and the cap give the converse inequality. This proves the side comparison. For x within $c\ell_S$ of S, the same comparison holds with ℓ_S on the left. The upper comparison follows by the Lipschitz property of d_Q and the size floor. For the lower comparison use the clearance when S is a square and the size floor otherwise. Intersecting neighborhoods are comparable as well. For example, if $\ell_T \le \ell_S$, any point of the intersection lies within $c\ell_S$ of S and within $c\ell_T$ of T, so both sides are comparable to the same clipped boundary distance. The area bound and disjointness now give the overlap estimates. If $l < l'$ occurs, a selected square of side l cannot have a parent failing the cap, since $2l \le l' \le \theta D$. Thus it lies in a boundary strip of width Cl, as do the terminal children. The area of this strip in a ball of radius $C'l'$ is $O(l'l)$, by covering it with the four strips along the sides of Q. Dividing by the child area proves the count. A fixed enlargement of a child only enlarges the radius of the ball used in the count. □

With a fixed small cap, the nonterminal child side contracts by an arbitrarily small fixed factor at each step. The old square is a shape-regular parallelogram in the new chart by the relative metric bound. Thus the comparison of its diameter with its old side is uniform. If the subdivision uses minimum side Rs and then a uniform $R \times R$ split of each nonterminal base square, the nonterminal contraction is at most C/R. In the fine split all children have their common side and no boundary clearance is required. Base terminal cells and unsplit parents have diameter $O(Rs)$ in the new chart. After a fixed-cap split, unsplit and merged terminal cells instead have diameter $O(s)$. All children that are retained in the partition have area $\gtrsim s^2$. If sides contract by σ, a descendant j stages later has diameter at most $C(\sqrt{K_0}\sigma)^j\ell_S$ in the chart of its ancestor S. This follows by composing the relative chart bounds. It permits the ancestor oscillations controlled by scaled Hölder seminorms to be summed geometrically by choosing $\sqrt{K_0}\sigma < 1$.

Let η_Q be a smooth probability density supported in a concentric subball of the inball, with $\|\eta_Q\|_\infty \lesssim D^{-2}$ and $\|\nabla\eta_Q\|_\infty \lesssim D^{-3}$. The ray kernel from source y is

$$K_Q(x,y) = (x-y)\int_1^\infty t\,\eta_Q(y+t(x-y))\,dt.$$

Indeed, send a unit source at y along the straight segment to z, averaged with $\eta_Q(z)\,dz$. The weak gradient pairing is $\int \eta_Q(z)\phi(z)\,dz - \phi(y)$, so the divergence is $\delta_y - \eta_Q$ on the whole plane, with the current extended by zero off Q. The kernel formula follows from $x = y + (z-y)/t$. Thus the ray inverse is exact for mean-zero sources, including the boundary condition. Direct differentiation gives, away from $x = y$,

$$\begin{aligned} |K_Q(x,y)| &\lesssim |x-y|^{-1}, \\ |\nabla_x K_Q(x,y)| + |\nabla_y K_Q(x,y)| &\lesssim |x-y|^{-2}, \\ |(\nabla_x + \nabla_y)K_Q(x,y)| &\lesssim (D|x-y|)^{-1}. \end{aligned}$$

For these estimates use $t|x-y| \le D$ in the integral. In the last derivative the argument of η_Q has simultaneous translation derivative one, and the prefactor has derivative zero. The last estimate

is valid distributionally as well, by differentiating under simultaneous translation with $x - y$ fixed. There is also the support bound

$$K_Q(x, y) = 0 \quad \text{when } |x - y| > C d_Q(x). \tag{2.1}$$

When the integrand is nonzero, $x = (1 - t^{-1})y + t^{-1}z$ with z in the central ball. A nearest supporting line has distance at least cD from z and nonnegative distance from y, so $d_Q(x) \geq cD/t$. In particular $\int |K_Q(x, y)|\, dy \lesssim d_Q(x)$. The two-pole difference bound $\int |K_Q(x, y) - K_Q(x', y)|\, dy \lesssim d(1 + \log(D/d))$, $d = |x - x'| \leq D/2$, follows by integrating the singular bound within distance $2d$ of the points and the derivative bound farther away along their connecting segment. This gives a continuous representative of the current of a bounded source, vanishing on ∂Q.

Lemma 3 (Coarse Gaussian current and envelopes)**.** *For the fixed-cap Whitney split, let P be child averaging and P_Q parent averaging. For $x, x' \in S^+$, the kernels $PK_Q(x, \cdot)$ and*

$$\left(\frac{\ell_S}{|x - x'|}\right)^{\beta} P(K_Q(x, \cdot) - K_Q(x', \cdot)), \qquad 0 < \beta < 1,$$

are supported in a ball of radius $C\ell_S$ centered at a representative of S, and have supremum at most C_β/ℓ_S. The same holds for $\ell_S P(x, \cdot)$, the normalized density test, where $P(x, \cdot) = \mathbf{1}_T/|T|$ when $x \in T$. Consequently the contrast current $V(x) = W((P - P_Q)K_Q(x, \cdot))$ has pointwise covariance bounded by CI. If $f = (P - P_Q)W$, then for each fixed $0 < \alpha < 1$ the even envelope

$$E_S = \sup_{S^+} |V| + \ell_S^{\alpha}[V]_{C^\alpha(S^+)} + \ell_S \|f\|_{L^\infty(S^+)}$$

has conditional tail $\mathbb{P}(E_S > t \mid \text{past}) \leq Ce^{-ct^2}$. The same conclusions hold for any fixed upper cap, with constants depending on that cap.

Proof. For fixed cap, Lemma 2 gives $d_Q(x) \lesssim \ell_S$ on S^+. If S is a square, $d_Q(x) \gtrsim \ell_S$ there, by clearance. If it is terminal use the size floor. Thus any child meeting a small ball of radius $c'\ell_S$ about x has area $\gtrsim \ell_S^2$. Every child meeting $B(x, C\ell_S)$ has diameter $O(\ell_S)$, by the upper side comparison and the Lipschitz bound on distance. The support bound follows from (2.1). On near children, $\int_{B(x, C\ell_S)} |x - y|^{-1} dy = O(\ell_S)$ divided by the child area gives the desired supremum. Farther away use the pointwise bound. For the difference, put $d = |x - x'|$. When d is a small fraction of ℓ_S, integrating the two singularities in the $2d$ balls and the derivative elsewhere gives $O(d(1+\log(\ell_S/d)))$ over the near region. Divide by the near child area. For far children the derivative estimate gives $O(d/\ell_S^2)$ pointwise. Use $v(1 + \log(1/v)) \lesssim_\beta v^\beta$. When d is comparable to ℓ_S, the single-point estimates suffice. The density bound follows from the neighborhood comparability. The resulting L^2 kernel bounds are uniform. Subtracting P_Q contracts L^2.

Here is the Gaussian envelope argument, also used for the fine split below. For any exponent β between α and one the L^2 bounds control the canonical increment distance by $C(|x - x'|/\ell_S)^\beta$. In the rescaled neighborhood, take nets of mesh 2^{-j} and cardinality at most $C2^{2j}$. Compare points in two successive nets within distance $C2^{-j}$, and also compare neighboring points at each net level. The expected maximum of the absolute Gaussian differences of these pairs is at most $C2^{-j\beta}\sqrt{1 + j}$, by the Gaussian tail and a union bound. Telescoping from the level of the separation of two points bounds the expected scaled α-Hölder seminorm by the convergent sum $C \sum_{j \geq 0} 2^{-j(\beta-\alpha)}\sqrt{1 + j}$. It also constructs a continuous version. The expected supremum is bounded by the same sum and the expected absolute value at one point. The density tests are constant on only a bounded number of cells in the neighborhood. Finally the three envelope terms are suprema of linear Gaussian tests,

with signs or unit vector directions included. Their test norms are uniformly bounded in L^2, so the sum is Lipschitz in the Gaussian coefficients with bounded constant. Gaussian concentration proves the tail, and the definition gives evenness. □

Lemma 4 (Sharp covariance for a uniform fine split)**.** *Let Q be a square subdivided uniformly into $R \times R$ squares of side ℓ. There is a zero-normal inverse for the fresh contrast with chart covariance at most*

$$\left(\sqrt{\frac{\log R}{4\pi}} + C\right)^2 I.$$

For x, x' in a fine child or its small ℓ-neighborhood inside Q, the canonical L^2 distance of the two evaluations is at most $C_\beta(|x - x'|/\ell)^\beta$ for every fixed $\beta < 1$. An even envelope controlling the supremum and scaled α-Hölder seminorm of the current, divided by $\sqrt{1 + \log R}$, has a uniform Gaussian tail.

Proof. Choose a smooth radial probability density ζ supported in the annulus $1/2 < |z| < 1$. For $y \in Q$, put $a_y = c\, d_Q(y)$ for a small fixed $c > 0$ and set $\zeta_y(z) = a_y^{-2}\zeta((z - y)/a_y)$. Define

$$R_y(x) = \frac{x - y}{2\pi|x - y|^2}\left(1 - \int_{B(y,|x-y|)} \zeta_y(z)\, dz\right),$$
$$B(x, y) = \int_Q K_Q(x, z)\zeta_y(z)\, dz, \qquad \widetilde{K}_Q(x, y) = R_y(x) + B(x, y).$$

Radial integration shows that R_y has divergence $\delta_y - \zeta_y$. Its support is in $B(y, a_y)$. The divergence of B is $\zeta_y - \eta_Q$, by the ray identity. Thus $\widetilde{K}_Q$ has divergence $\delta_y - \eta_Q$ and zero normal flux. Both summands vanish for $|x - y| > Cd_Q(x)$. For B this follows from the ray support bound and $a_y \le c(d_Q(x) + |x - y|)$. The same inequality and the radial support give the assertion for R_y.

On this support B and $\nabla_x B$ are bounded by $C/d_Q(x)$ and $C/d_Q(x)^2$, respectively. When $|x-y| \ge d_Q(x)/4$, the smallness of c keeps the support of ζ_y a distance comparable to $|x-y|$ from x, so the estimates follow from the ray bounds. Otherwise $d_Q(y) \asymp d_Q(x)$, and integration of $|x - z|^{-1}$ against ζ_y gives the first bound. For the derivative, replace $\nabla_x K_Q$ by $(\nabla_x + \nabla_z)K_Q - \nabla_z K_Q$ and integrate the second term by parts against ζ_y. The simultaneous derivative is bounded by $C/(D|x - z|)$, and the other term uses $\|\nabla\zeta_y\|_\infty \lesssim d_Q(x)^{-3}$. This proves the second bound. The radial term and its derivative are bounded by $C/|x - y|$ and $C/|x - y|^2$. It follows that the full kernel satisfies the same pointwise bounds. Its L^1 norm in the source variable is $O(d_Q(x))$, by support and the singular bound, so currents of bounded sources extend continuously by zero on ∂Q.

Let P now average on the uniform fine grid and fix a unit direction e. On $|x - y| \le C_1\ell$, the projected singular kernel has L^2 norm $O(1)$. Indeed only $O(1)$ fine cells meet that disk, their area is ℓ^2, and the L^1 norm in the disk is $O(\ell)$. The entire regular part $B(x, \cdot)$ has L^2 norm $O(1)$ by its support and pointwise bound. On $C_1\ell < |x - y| < c_1 d_Q(x)$, with c_1 sufficiently small, the radial kernel is the free-space vector $(x - y)/(2\pi|x - y|^2)$. This is because $d_Q(y) \asymp d_Q(x)$ there and ζ_y vanishes on $B(y, a_y/2)$. The whole annulus lies inside Q. Its squared L^2 norm in direction e is

$$\int_{C_1\ell}^{c_1 d_Q(x)} \int_0^{2\pi} \frac{\cos^2\vartheta}{4\pi^2 r^2} r\, d\vartheta\, dr = \frac{1}{4\pi}\log\left(\frac{c_1 d_Q(x)}{C_1\ell}\right).$$

The remaining radial support lies in an annulus with bounded ratio of outer and inner radii, giving L^2 norm $O(1)$. If the displayed annulus is empty the bounds on the near part and remaining support

still give $O(1)$. Orthogonal projection contracts L^2, and $d_Q(x)/\ell \le CR$. The triangle inequality for these pieces therefore gives the stated covariance for $W((P-P_Q)\widetilde{K}_Q(x,\cdot))$.

For the difference let $d = |x - x'| \le c'\ell$. On a disk of radius $C_1\ell$ containing the two poles, the two-pole L^1 bound is $O(d(1+\log(\ell/d)))$. Averaging on the $O(1)$ intersecting cells gives L^2 norm bounded by this quantity divided by ℓ. Outside the disk the derivative bound gives squared L^2 norm $O(d^2/\ell^2)$ before projection. For d comparable to ℓ the same split uses the separate L^1 singular bounds on the near region and the derivative bound on its complement. This proves the canonical increment estimate. The net argument of Lemma 3 now gives an expected oscillation envelope $O(1)$ before division, and the expected supremum is $O(\sqrt{1+\log R})$ by adding one evaluation. The normalized linear tests have uniformly bounded L^2 norm, so Gaussian concentration and evenness hold as claimed. □

3 Local flow estimates and repair of omitted sources

3.1 Uniform capacity on a square mesh

Let D be a square made of K^2 grid cells, with K a power of two. Orient the internal grid edges and let dq denote the outgoing load minus the incoming load at each cell. Put

$$\mathcal{B}_D(g) = \min_{dq=g} \max_e |q_e|, \qquad \sum_v g_v = 0.$$

There are no loads on the exterior boundary. For a cell set S write ∂S for the internal edges between S and its complement. Then

$$\mathcal{B}_D(g) = \max_{\varnothing \ne S \ne D} \frac{|\sum_{v\in S} g_v|}{|\partial S|}. \tag{3.1}$$

The upper bound in (3.1) follows by separation of g from the divergence of the edge cube of the proposed capacity. Indeed, the support function of this image at a vector u is the capacity times $\sum_{v\sim w} |u_v - u_w|$. Subtract $\min_v u_v$ from u and integrate its superlevel sets. The integral of their cut sizes is $\sum_{v\sim w} |u_v - u_w|$, and the integral of their signed g-masses is $\sum_v u_v g_v$. Thus the cut inequalities give every support-function inequality for the image. The converse is obtained by summing $dq = g$ over S.

For $|S| \le K^2/2$, if no row or column is full, the product of the numbers of occupied rows and columns is at least $|S|$. Each occupied row or column contains a cut edge in its own direction. If a row is full, every column meets S and at least $K/2$ columns are mixed, and the argument for a full column is the same with axes exchanged. Hence

$$|\partial S| \ge c\sqrt{\min(|S|, |D \setminus S|)}.$$

It follows from Cauchy–Schwarz and (3.1) that $\mathcal{B}_D(g) \le C\|g\|_2$, and the triangle inequality gives the same Lipschitz bound for $\mathcal{B}_D$.

When the cell width is w, linearly interpolate each component between its two normal faces inside each cell, leaving it constant in the other coordinate. The resulting current has component suprema at most the maximum face load, divergence g_v/w on cell v, and zero normal trace on the box boundary. The normal traces agree across internal faces. A separate coordinate primitive will also be useful. In a rectangle $I \times J$ with a bounded mean-zero density f, put $r(y) = \int_I f(x,y)\,dx$ and use the two components

$$J_x(x,y) = \int_{\inf I}^{x} \left(f(t,y) - \frac{r(y)}{|I|} \right) dt, \qquad J_y(x,y) = \frac{1}{|I|} \int_{\inf J}^{y} r(t)\,dt. \tag{3.2}$$

These have divergence f and zero outer normal trace. In particular, on a square of side w both bounds are $2w\|f\|_\infty$.

Lemma 5. *Let G be a centered Gaussian mean-zero cell vector in D with covariance at most the orthogonal projection P onto mean-zero vectors. Then*

$$\mathbb{E}\mathcal{B}_D(G) \le C(1+\log K), \qquad \mathbb{P}\{\mathcal{B}_D(G) > C(1+\log K)+u\} \le e^{-cu^2}.$$

Proof. For independent standard Gaussian cell coefficients Z, telescope PZ into dyadic child-average contrasts. In each parent of side l grid cells the four child values have standard deviations at most C/l. Route their contrast by row partial sums after removing the row means, then use a vertical partial sum of the row means. The parent is balanced, so both outer normal loads vanish. Each face load is linear in the four child values, with coefficients bounded by Cl, and therefore has variance at most C. The levels of the dyadic contrast are independent by orthogonality. Summing the linear flows gives face variance at most $C(1+\log K)$, and the Gaussian tail bound on the $O(K^2)$ faces gives expected maximum at most $C(1+\log K)$. For covariance dominated by P, add an independent centered Gaussian of the covariance difference and apply convexity to get the same mean bound. The Euclidean Lipschitz bound from (3.1) and Gaussian concentration give the tail. □

We use Gaussian concentration in its Lipschitz form

$$\mathbb{E}e^{t(F(Z)-\mathbb{E}F(Z))} \le e^{t^2L^2/2}$$

for an L-Lipschitz function F of a standard Gaussian vector. One derivation is to set $P_tg(x) = \mathbb{E}g(e^{-t}x+\sqrt{1-e^{-2t}}Z)$. Gaussian integration by parts gives $-\partial_t\mathbb{E}[(P_tg)\log(P_tg)] = \mathbb{E}[|\nabla P_tg|^2/P_tg]$. Commuting the gradient through P_t and applying Cauchy–Schwarz bounds the right side by $e^{-2t}\mathbb{E}(|\nabla g|^2/g)$. Integrating in t gives the Gaussian entropy bound with constant $1/2$. With $g = e^{sF}$ it bounds $\partial_s(s^{-1}\log\mathbb{E}e^{sF})$ by $L^2/2$, which gives the displayed moment bound upon integration. Approximation extends the calculation from smooth bounded functions to Lipschitz F.

Lemma 6. *Let $g = \sum_i \sigma_i(e_{V_i} - p_i)$, where the V_i are independent cell labels, $p_i = \mathbb{E}e_{V_i}$, and $\sigma_i \in \{-1,1\}$ are fixed. If the total mean occupancy per cell is at most $C\mu$, $\mu \asymp m$, and $K \le m^C$, then*

$$\mathbb{E}\mathcal{B}_D(g) \le C'\sqrt{\mu}(1+\log K), \qquad \mathbb{P}\{\mathcal{B}_D(g) > C_A\mu\} \le e^{-Am}$$

for each fixed A, with the constant chosen accordingly. These estimates also apply conditionally with the same parameters.

Proof. Use an independent replica g' and Jensen to bound the mean by that of $g-g'$. The paired label differences $\sigma_i(e_{V_i} - e_{V_i'})$ can be symmetrized by independent signs, since the ordered pairs are exchangeable. Given the two label arrays, replace the signs by standard normals at a constant dilation $1/\mathbb{E}|Z_1|$. This follows by conditioning on the Gaussian signs and applying Jensen over their absolute values. If D_* is the largest pooled occupancy of a cell in the two arrays, the conditional covariance of this Gaussian sum is at most CD_*P, by $\sum_i(u_{V_i} - u_{V_i'})^2 \le 2D_*\|Pu\|_2^2$. Each pooled cell count N_v has $\mathbb{E}e^{N_v} \le \exp(C'\mu)$ by the product moment of its independent indicators. Thus $\mathbb{E}D_* \le C'\mu + \log(K^2)$, and $\mathbb{P}(D_* > C'_A\mu) \le e^{-2Am}$ after enlarging the constant. The conditional Gaussian mean bound and Jensen give the claimed unconditional mean. On $D_* \le C'_A\mu$, apply the sign and covariance comparisons to the convex function $\exp(t\mathcal{B}_D)$. The Gaussian moment bound gives

$$\exp\bigl(C''_At\sqrt{\mu}(1+\log K) + C''_At^2\mu\bigr).$$

A fixed positive tilt and the pooled occupancy estimate give $\mathbb{P}(\mathcal{B}_D(g-g') > C'''_A\mu) \le \frac{1}{2}e^{-Am}$ for a suitable constant. Finally for $u = C'''_A\mu$ the probability that $\mathcal{B}_D(g') \le u$ is at least $1/2$ by the mean bound. Independence and the triangle inequality then give the asserted bound for $\mathcal{B}_D(g) > 2u$. □

Here and below we use the following elementary count estimates. For a sum S of independent indicators of total mean μ, its moment generating function is bounded by $\mathbb{E}e^{t(S-\mu)} \le \exp(\mu(e^t-1-t))$. This is exact for a Poisson variable of mean μ. For $u \ge 0$, exponential tilts of both signs give $\mathbb{P}(|S-\mu| > u) \le 2e^{-c\min(u^2/\mu,u)}$. For example, use $e^t - 1 - t \le Ct^2$ for $|t| \le 1$ and choose $|t|$ a sufficiently small constant times $\min(1, u/\mu)$. The same argument applied to independent centered random variables X_i with $|X_i| \le B$ gives $\mathbb{P}(|\sum_i X_i| > u) \le 2e^{-c\min(u^2/v,u/B)}$ for $v = \sum_i \mathbb{E}X_i^2$, by using $e^{tX_i} - 1 - tX_i \le Ct^2X_i^2$ for $|t| \le 1/B$. In particular the estimates apply to each cell in a multinomial count, and conditionally to any independent labels left by conditioning on the coarser counts.

3.2 Joint tails for coarse envelopes

At a predictable generation let S range over the Whitney children of the old parents. Write l_S for intrinsic child size and E_S for the coarse envelope in the parent chart. We use the local kernel and packing bounds of Lemmas 2 and 3. In particular the envelope is a sum of a fixed number of suprema of absolute normalized Gaussian tests, each with a kernel of size at most C/l_S and supported within Cl_S of S, before the projection away from parent constants. The single-child mean is bounded uniformly.

Lemma 7. *Conditionally on the old history, for each predictable finite subset $\mathcal{A}$ of attempted children and sufficiently large v,*

$$\mathbb{P}\{E_S > v \text{ for all } S \in \mathcal{A} \mid \text{old history}\} \le \exp(-c|\mathcal{A}|v^2).$$

Proof. Consider one normalized test kernel k_S selected for each $S \in \mathcal{A}$. For overlapping supports in the same chart with $l \le l'$ their scalar product is at most Cl/l' in absolute value, by the support-area bound. For a fixed child of size l', the number of overlapping kernels with size $l < l'$ is $O(l'/l)$. Indeed their disjoint child areas lie in a boundary strip of width Cl within a ball of radius Cl'. For each $l' \ge l$, at most $O(1)$ of the larger supports can overlap a fixed size-l support, by disjoint areas comparable to l'^2. Weighted Schur with weights l_S^η, $0 < \eta < 1$, bounds the Gram matrix uniformly. In the sum from smaller children the factor is $(l/l')^\eta$, and in the sum from larger children it is $(l/l')^{1-\eta}$. The resulting geometric sums are finite. The same argument for comparable scales is just area packing. Thus for arbitrary coefficients a_S, $\|\sum_S a_S k_S\|_2^2 \le C\sum_S |a_S|^2$. Within each parent projection away from its constant is a common orthogonal projection, which decreases this norm. The projected tests from distinct parents have disjoint supports. Taking suprema over the choices of one signed test per child and summing a bounded number of these suprema, $\sum_{\mathcal{A}} E_S$ is consequently $C\sqrt{|\mathcal{A}|}$-Lipschitz. Its conditional mean is $O(|\mathcal{A}|)$. Gaussian concentration at $|\mathcal{A}|v$ proves the result. □

3.3 Coarse clipping and exact sparse repair

Fix a threshold E_0. Take a Lipschitz cutoff χ_S that is one on S and is supported in the small chart neighborhood defining E_S, with $|\nabla\chi_S| \le C/l_S$ in that chart. For example, use the positive part of $1 - \operatorname{dist}(\cdot, S)/(cl_S)$ with c sufficiently small. Call S exceptional if $E_S > E_0$ and multiply its parent's whole current by

$$1 - \chi_* = \prod_{S \text{ exceptional}} (1 - \chi_S).$$

This multiplier is even in the new noise and is between zero and one. Bounded comparable overlap implies that on each nonexceptional child the supremum and scaled Hölder bounds for the retained current are CE_S, and on exceptional children the retained current is zero. To verify the Hölder bound within a child, use the product difference inequality for factors in $[0,1]$, the uniformly bounded number of cutoffs meeting the child, and their comparable $O(1/l_S)$ gradient bounds. These are the bounds on the child used in the evolution. At any fixed point the even multiplier only decreases the conditional linear exponential moment, by averaging over the signs of the entire new Gaussian vector and using that $\cosh(v)$ increases with $|v|$.

Order the exceptional children within a parent and decompose $\chi_* V = \sum_S \varphi_S V$, with $\varphi_S = \chi_S \prod_{T<S}(1-\chi_T)$. For the current multiplied by a deterministic scale σ, the omitted source $f_S = \sigma \operatorname{div}(\varphi_S V)$ is bounded by $C\sigma E_S/l_S$ in its chart neighborhood. The product rule uses the density test in E_S and $|\nabla \varphi_S| \le C/l_S$, again by comparable overlap. The parent current has zero normal trace, so the product rule across its boundary introduces no measure on that boundary. Thus each source is a genuine bounded function of integral zero, supported in the physical image of a chart ball of radius Cl_S intersected with the parent.

Lemma 8. *Suppose $m = \log(1/h)$, there are at most Cm generations, all physical charts have determinant one and distortion bounded by a fixed power of m, and $h \le l_S \le m^C$. Assume the domain is an area-one rectangle with both sides and their reciprocals at most m^C. Take $\log m = o(E_0^2)$, $E_0^2 \le m$, and set*

$$D_0 = m/E_0^2, \qquad T_0 = \sigma\sqrt{m}.$$

For any $1 \le B \le m^C$, with probability $1-o(1)$ the sum of all omitted sources has an exact current with zero outer normal trace, supported on a set of area at most $\exp(-cE_0^2)$, and satisfying

$$\|J_y\|_\infty \le CT_0 D_0/B, \qquad \|J_x\|_{\psi_2} \le CT_0 D_0 B/E_0.$$

Proof. A conditional single-child tail and a union bound over the at most Cmh^{-2} attempted children up to constants give $E_S \le C\sqrt{m}$ everywhere with probability $1-o(1)$. Thus the source in its support ellipse satisfies

$$\|f_S\|_\infty \le CT_0/l_S, \qquad \|f_S\|_1 \le CT_0 l_S.$$

If A is the chart map and $G = AA^{\mathsf{T}}$, the horizontal and vertical maximal chord lengths of the ellipse are bounded by $Cl_S/\sqrt{G_{yy}}$ and $Cl_S/\sqrt{G_{xx}}$. This follows by imposing zero orthogonal physical coordinate on a chord in the equation $(z-z_0)^{\mathsf{T}} G^{-1}(z-z_0) \le Cl_S^2$ and using $\det G = 1$. Consequently the absolute horizontal and vertical line masses A_x, A_y of f_S satisfy $A_x A_y \le CT_0^2$.

Around each candidate ellipse choose a predictable axis box inside the domain with both sides at least Bl_S. For instance enclose the ellipse portion in the domain in its axis bounds, enlarge each side by Bl_S as necessary, and slide the intervals inside the domain. The side requirements are at most $m^{C'} l_S$ by the chart cap. If the box cannot fit, declare failure only if S is exceptional. Such a candidate has size at least $m^{-C'}$, so there are only polynomially many per generation by area packing. The probability of a failure is $o(1)$ by the conditional envelope tail.

When $A_x \le T_0 B$, apply (3.2) in the candidate box. Its horizontal supremum is at most $2A_x$ and its vertical one at most $\|f_S\|_1/|I| \le CT_0/B$. If $A_x > T_0 B$, then $A_y \le CT_0/B$, and applying the column version gives the same bounds with a changed constant, since the horizontal supremum of the spread mass primitive is at most $CT_0/B \le CT_0 B$. Extend the box currents by zero. Their zero normal traces preserve the divergence.

To bound overlap, fix a point and pad the candidate boxes by $O(h^2)$ on all sides. The number n of these boxes containing the point in a generation is at most $m^{C''}$. In each intrinsic dyadic size class their disjoint child areas are comparable to the squared size and lie within polynomial times

that size of the point. The padding preserves this bound as $l_S \geq h$. If I_S is the exceptional indicator, expand

$$\mathbb{E}\exp\left(\theta\sum_S I_S\right) = \mathbb{E}\prod_S(1+(e^\theta-1)I_S) \leq \exp(ne^{-cE_0^2/2}), \qquad \theta = cE_0^2/2,$$

conditionally on the old history, by Lemma 7. Iterate through the generations. The probability of overlap larger than $C'D_0$ at the point is at most $\exp(-c'C'm + o(1))$. A union bound over a mesh of spacing h^2 and the padding give a uniform cap $O(D_0)$ for a sufficiently large C'. The sum of all candidate box areas in a generation is bounded by a fixed power of m by disjoint child areas. The expected total area of exceptional boxes is thus $m^{C'''}e^{-cE_0^2}$, by the single envelope tail. Markov's inequality gives the stated area bound with high probability. A function bounded by M on a set of area p and zero off it has Luxemburg norm at most $M/\sqrt{\log(1+1/p)}$. Apply this with $M = CT_0D_0B$ and the area bound. The overlap cap also gives the vertical supremum bound. □

4 The sharp Gaussian lower bound

Set $m = \log(1/h)$, and let R be a dyadic integer with $\log R \sim (\log m)^2$. We construct a random Lipschitz test potential u by adding local Poisson solutions on successively smaller squares. Its vertical derivative is positive with integral one. Changing spatial measure by this derivative makes the ratio of horizontal to vertical derivatives a martingale. We control that ratio by stopping stages of a state-dependent variance clock.

4.1 One Poisson refinement

On $S = (-1/2, 1/2)^2$, subdivide the lattice squares of side R^{-1} into four triangles by joining their corners to their centers. Let I_R be nodal affine interpolation. Take a dihedrally invariant smooth cutoff η which is zero on a boundary strip of width $m^{-A_0}/2$ and one outside the strip of width m^{-A_0}. Its derivatives of order k are bounded by $C_k m^{kA_0}$. Let ϕ_k run over an orthonormal real Fourier basis in the band

$$m^{A_1} \leq |k| \leq Rm^{-A_1}, \qquad \lambda_k = 4\pi^2|k|^2,$$

where both sine and cosine are included for each frequency pair, and A_1 is a sufficiently large fixed constant in terms of A_0. Denote these cutoffs by $K_{\min}$ and $K_{\max}$. Put $g_k = (I - \Pi_{\text{aff}})I_R\phi_k$, where Π_{aff} is orthogonal projection onto affine functions. Define f_k by subtracting from $I_R(\eta\phi_k)$ a linear combination of $I_R(\eta_0 p)$, $p = 1, x_1, x_2$, so that f_k is orthogonal to all affine functions. Here η_0 is a fixed dihedrally invariant nonnegative smooth cutoff supported strictly inside S and positive on an open subset. The 3×3 moment matrix converges to the positive definite Gram matrix with weight η_0. This defines the correction uniquely for large m, and it commutes with the dihedral action. Both g_k and f_k are piecewise affine, and f_k vanishes near ∂S.

For white noise $\mathcal{W}$ on S, let

$$F = \sum_k \lambda_k^{-1}\mathcal{W}(g_k)f_k, \qquad V_R = \frac{\log R}{4\pi}.$$

The coefficient covariance matrix is $I + o(1)$ in operator norm. Indeed, on each shape-regular triangle the interpolation error in L^2 is at most CR^{-2} times the L^2 norm of the second derivatives. To obtain the unit-scale estimate, subtract the affine function making the mean and mean first derivatives zero, apply Poincaré twice, and use the embedding of H^2 in C^0 in two dimensions to

control the nodal values. Summing over triangles gives an error operator of norm $O((K_{\max}/R)^2)$ on the Fourier band. The affine projection of the Fourier band has norm $O(K_{\min}^{-1/2})$ because the Fourier coefficients of each coordinate lie on its coordinate axis and are $O(|k|^{-1})$, whose squares sum to $O(K_{\min}^{-1})$. These two estimates give the claimed Gram matrix bound.

Away from triangle boundaries, the weighted coefficient vector of ∇F differs in ℓ^2 by $o(1)$ from the vector $(\lambda_k^{-1}\eta\nabla\phi_k)_k$. The interpolation gradient error in a single frequency is bounded by CR^{-1} times the supremum of its second derivatives, by Taylor's formula at the triangle vertices. Here that gives

$$|\nabla I_R(\eta\phi_k) - \nabla(\eta\phi_k)| \le CR^{-1}(|k|^2 + m^{A_0}|k| + m^{2A_0}).$$

The weighted ℓ^2 error is $O(K_{\max}/R)$ as $K_{\min} \gg m^{2A_0}$. The term containing $\nabla\eta$ costs $O(m^{A_0}/K_{\min})$ in the same norm. The moment corrections have uniformly bounded coefficients since the uncorrected output functions are uniformly bounded and the moment matrix has bounded inverse. Their weighted ℓ^2 gradient contribution is $O(K_{\min}^{-1})$. The same estimates imply $\|\nabla f_k\|_2 \le C|k|$ and $\|f_k\|_2 \le C$. In the ideal periodic gradient, the sine-cosine pair sums and lattice symmetry give

$$\sum_k \lambda_k^{-2}\nabla\phi_k(x)\nabla\phi_k(x)^{\mathsf{T}} = \left(\frac{1}{4\pi}\log(K_{\max}/K_{\min}) + O(1)\right) I.$$

In fact the trace equals $(4\pi^2)^{-1}\sum_{K_{\min}\le|k|\le K_{\max}}|k|^{-2}$ with the sum over lattice vectors. Comparing with the radial integral gives $2\pi\log(K_{\max}/K_{\min}) + O(1)$ for the lattice sum, since the derivative of $|x|^{-2}$ is integrable away from the origin. Thus the covariance of ∇F is bounded by $CV_R I$ and is $(1+o(1))V_R I$ outside the boundary strip. The integral in x of the covariance is a scalar matrix by dihedral symmetry.

The expected unprojected pairing is

$$\mathbb{E}\sum_k \lambda_k^{-1}\mathcal{W}(g_k)\mathcal{W}(f_k) = \sum_k \lambda_k^{-1}\langle g_k, f_k\rangle = (2+o(1))V_R.$$

For the last equality, $\langle g_k, f_k\rangle = \langle\phi_k, \eta\phi_k\rangle + o(1)$ uniformly. For the interpolation errors use $K_{\max}/R = o(1)$ and the preceding bounds. For the moment correction, each $I_R(\eta_0 p)$ has inner product $o(1)$ with each band mode by integration by parts and its $O(R^{-2})$ interpolation error. Summing the sine-cosine pairs replaces $\langle\phi_k, \eta\phi_k\rangle$ by $\int_S \eta = 1 + o(1)$. The same lattice sum proves the formula.

Clip F to zero if $\|\nabla F\|_\infty > C\log R$. This event has probability $O(R^{-10})$ with a sufficiently large fixed C, by a Gaussian tail and a union bound over the $4R^2$ triangles. Let $\widehat{F}$ be the clipped function. Its integrated covariance is still a scalar matrix vI, since the clipping event is invariant under the dihedral group. Gaussian fourth moments and Cauchy–Schwarz show $v = (1+o(1))V_R$. Set $U = \widehat{F}/\sqrt{v}$. For an independent uniform point of S, the gradient $\xi = \nabla U$ is centered with covariance I and satisfies

$$|\xi| \le C\sqrt{\log R}, \qquad \int_S \nabla U = 0 \quad \text{for every realization.} \tag{4.1}$$

The clipped pairing has expectation $(1+o(1))\sqrt{\log R/\pi}$. Indeed, the unconditional L^2 norm of the pairing before division by $\sqrt{v}$ is $O(\log R)$, by Gaussian fourth moments of each factor, Minkowski, and $\sum_k \lambda_k^{-1} = O(\log R)$. Cauchy–Schwarz bounds the clipping error by $o(\log R)$.

4.2 The tilted stage calculation

Fix constants $\gamma, B > 0$ and $0 < \varepsilon < 1$. Put $\kappa = 1/20$ and choose

$$J = \left\lfloor \frac{\kappa \log m}{8 \log\log m} \right\rfloor, \quad q = \frac{\gamma}{J}, \quad L = \log(1/q), \quad S_j = m^{\kappa} L^{4j}, \quad b_j = \frac{B}{\sqrt{S_j}}, \quad \ell_j = q b_j, \quad A_j = \frac{S_j}{2L}.$$

Then $m^{\kappa} \le S_j \le m^{2\kappa}$ for large m, and $L \sim \log\log m$. Let $n \sim m/\log R$ be the maximum number of refinement steps, fixed below. In stage j take

$$a_j(x) = \frac{A_j}{x^2 + \ell_j^2}, \qquad r_j = \frac{1}{n\lambda_j}, \qquad du = r_j a_j(x)^{-1/2}, \qquad ds = r_j a_j(x)^{1/2}, \tag{4.2}$$

where $\lambda_j \asymp_{\gamma,B,\varepsilon} L^{-1/2}$ is specified below. These are the variance clock increments and $\sqrt{du\, ds} = r_j$ is the resource paid at each step.

Let (X, Y) denote the gradient of the current potential on a branch, initially $(0, 1)$, and write $M = X/Y$. In stage j, the coordinate x in (4.2) is M minus its value at the entrance to the stage. On an active step use

$$\begin{pmatrix} \Delta X \\ \Delta Y \end{pmatrix} = Y\sqrt{r_j} \begin{pmatrix} a_j^{-1/4} & M a_j^{1/4} \\ 0 & a_j^{1/4} \end{pmatrix} \xi.$$

Writing $\beta = \Delta Y / Y$ and $c = (\Delta X - M\Delta Y)/Y$, the ordinary centered innovations have variances ds and du, respectively. Their absolute values are bounded by $C\sqrt{ds \log R}$ and $C\sqrt{du \log R}$. Each updated block preserves the integral of Y, by (4.1). The pathwise spatial law tilted by the final Y is a probability measure because the initial integral is one. We write ν for the joint law obtained by applying this tilt to the uniform spatial branch for each noise realization.

On a retained block, the conditional change of measure is $1 + \beta$. To see this even with geometry chosen adaptively, condition on the earlier reveals and the old branch cell. The integral of the final Y over any current cell equals its current constant Y times its area, since every later perturbation lies inside a subcell and has zero integrated gradient. If a retiling discards an area fraction θ of the cell, that region keeps the old Y, and the retained squares each have spatial average Y. The discard probability is thus exactly θ under either law, conditional on the old data. Conditional on retention, the branch is uniform in its new square before the next innovation. For each fixed fresh field, its total future spatial weight in that square is the old Y times the area. The marginal fresh Gaussian field therefore retains its original law, and the child selection density relative to uniform is $1 + \beta$. Consequently

$$\Delta M = \frac{c}{1+\beta}, \qquad \mathbb{E}_\nu[\Delta M \mid \text{past}] = 0, \qquad \mathbb{E}_\nu[(\Delta M)^2 \mid \text{past}] = (1 + o(1))du$$

on retained steps. In fact the last conditional expectation equals $\mathbb{E}[c^2/(1+\beta) \mid \text{past}]$ under the original law. Before $|x|$ exits b_j, (4.2) gives

$$ds \le CS_j/(qn), \qquad du/b_j^2 \le CL/n, \qquad |\beta| \le C\sqrt{S_j \log R/(qn)} = O(m^{-c_0})$$

for an absolute $c_0 > 0$. The ratio jumps are bounded by $D_j = Cb_j\sqrt{L \log R/n}$. Uniformly in j, $D_j/\ell_j = O(m^{-c_0})$ and $\sum_j D_j = o(b_J/\sqrt{\log J})$, by decreasing c_0 if necessary. These bounds also hold for synthetic continuations using the same law of ξ, tilted by $1 + \beta$ in each step.

For the moment consider an untruncated stage, with no time, geometry or clock cap. Its relative ratio x_k starts at zero, and the discrete local time is

$$L_k^0 = |x_k| - \sum_{i=1}^{k} \operatorname{sgn}_+(x_{i-1})(x_i - x_{i-1}), \qquad \operatorname{sgn}_+(0) = 1.$$

Its increments are nonnegative and at most $2D_j$, by the convexity and Lipschitz bounds for $|x|$. A positive increment ends within D_j of zero. Stop the stage on $L_k^0 \geq \ell_j$ (a success) or $|x_k| \geq b_j$ (a failure F_j). These conditions cannot occur in the same step for large m. The stopped position is bounded by $b_j + D_j$, and the conditional variance before stopping is bounded below by a positive deterministic number for fixed m, j. Applying the stopped square identity at bounded times shows that the stopping depth has finite mean. Optional stopping for the Tanaka identity gives $\mathbb{E}_\nu|x_T| = \mathbb{E}_\nu L_T^0$. Thus the conditional failure probability p_j at entry satisfies

$$p_j b_j \leq \ell_j + 2D_j, \qquad (1-p_j)\ell_j \leq p_j(b_j + D_j) + (1-p_j)D_j.$$

In particular $p_j = (1+o(1))q$ uniformly in the entry history. From an intermediate state x the same argument gives $\mathbb{E}_\nu[|x_T| \mid \text{past}] \leq |x| + \ell_j + 2D_j$.

Write b, ℓ, A, D for b_j, ℓ_j, A_j, D_j in the current stage. For $w_i(x) = (x^2+\ell^2)^{-i/2}$, $i = 1, 2$, let g_i be the even function with $g_i'' = w_i$, $g_i(0) = g_i'(0) = 0$. The logarithmic derivative of w_i is bounded by $2/\ell$. On a jump of size at most $D = o(\ell)$, the quadratic Taylor remainder for g_i is therefore $(1+o(1))w_i(x)(\Delta x)^2/2$, uniformly. Conditional means and the variance formula give

$$\mathbb{E}_\nu \sum_{k<T} w_i(x_k)\,du_k = (2+o(1))\mathbb{E}_\nu g_i(x_T). \tag{4.3}$$

The corresponding estimate starting from an intermediate state subtracts $g_i(x)$ on the right. One may first sum to bounded stopping times and then pass to the limit by boundedness of $g_i(x_{k\wedge T})$ and nonnegativity of the occupation sum. The two functions are

$$g_1(x) = |x| \operatorname{arsinh}(|x|/\ell) - \sqrt{x^2+\ell^2} + \ell, \qquad g_2(x) = (|x|/\ell)\arctan(|x|/\ell) - \tfrac{1}{2}\log(1 + x^2/\ell^2).$$

Since $g_1(b) = (1+o(1))bL$, $g_2(b) \leq Cb/\ell$, and $g_i(D) \leq CD^2/\ell^i$, (4.3) and the endpoint estimates give $\mathbb{E}_\nu \sum w_1\,du = (2+o(1))qbL$ and $\mathbb{E}_\nu \sum w_2\,du = O(1)$. For the total resource $R_j = \sqrt{A}\sum w_1\,du$ this proves

$$\mathbb{E}_\nu[R_j \mid \text{entry}] = (1+o(1))\mu_j, \qquad \mu_j = \sqrt{2}Bq\sqrt{L}. \tag{4.4}$$

Also, conditional on any intermediate state x, the chance of eventual failure is at most $(|x|+\ell+2D)/b$. Multiplying by $ds = Aw_2(x)\,du$, summing predictable steps and using $|x|w_2(x) \leq w_1(x)$ gives

$$\mathbb{E}_\nu[s_j\, 1_{F_j} \mid \text{entry}] \leq \frac{A}{b}\,\mathbb{E}_\nu \sum \big(w_1(x) + (\ell+2D)w_2(x)\big)\,du \leq (1+o(1))qS_j. \tag{4.5}$$

We also need bounds from arbitrary intermediate states. Convexity gives $g_i(z)/z$ nondecreasing for $z > 0$, so

$$\mathbb{E}_\nu[g_i(x_T) \mid \text{past}] - g_i(x) \leq \frac{g_i(b+D)}{b+D}(|x| + \ell + 2D) - g_i(x).$$

For $i = 1$ this is $O(b)$, since $g_1(z) = |z|\log(1+|z|/\ell) + O(|z|+\ell)$, $|x|\log((2b)/(|x|+\ell)) \leq Cb$, and $\ell L = o(b)$. For $i = 2$, use $g_2(z) = \pi|z|/(2\ell) - \log(1+|z|/\ell) + O(1)$ to bound it by $O(L)$. The conditional occupation bound thus gives $\mathbb{E}_\nu[R_{j,\text{future}} \mid \text{past}] \leq Cb\sqrt{A}$ and $\mathbb{E}_\nu[s_{j,\text{future}} \mid \text{past}] \leq CS_j$.

Writing R_j^2 as the sum of the squared resource steps plus twice the cross-products, and conditioning after the earlier step, gives

$$\mathbb{E}_\nu[R_j^2 \mid \text{entry}] \le (r_j + Cb\sqrt{A})\mathbb{E}_\nu[R_j \mid \text{entry}] = O(q). \tag{4.6}$$

For the clock tail, from each active state the probability of accumulating another KS_j units is at most $1/2$ for a fixed sufficiently large K, by conditional Markov. Restarting at these thresholds and using the per-step overshoot $o(S_j)$ gives the exponential tail. We cap each actual stage at $CS_j \log J$ with C large, so that the probability of hitting any cap is $O(J^{-5})$.

4.3 Choice of the time allocation

Let $p_j^0 = (1-q)^{j-1}$, and choose

$$T_0 = \sum_{j=1}^{J} (p_j^0)^2 \mu_j, \qquad \lambda_j = \frac{(1-\varepsilon)(p_j^0)^2}{T_0}.$$

Then

$$T_0/\sqrt{L} \;\longrightarrow\; \frac{B}{\sqrt{2}}(1 - e^{-2\gamma}), \qquad \lambda_j \asymp_{\gamma,B,\varepsilon} L^{-1/2}. \tag{4.7}$$

Each resource step in stage j contributes time $\lambda_j r_j = 1/n$. For the time comparison, continue the ideal process after the first failure, with synthetic restarts at zero, and append synthetic innovations if a real branch is lost or budget-capped. Use the same tilted single-step law in the appended part. It is defined for all admissible states by (4.1) and (4.2). The uniform conditional estimates above hold for every appended stage. The predictable means of $\sum_j \lambda_j R_j$ sum to $1 - \varepsilon + o(1)$ by (4.4), and the martingale part has variance $O(\sum_j \lambda_j^2 q) = O(1/L)$ by (4.6). Thus the chance of exhausting the budget is $O(1/L)$. Similarly the ideal reward $\sum_j \sqrt{\lambda_j} R_j$ has second moment $O(\sqrt{L})$ because its sum of conditional means is $O(L^{1/4})$, and its martingale part has variance $O(\sum_j \lambda_j q) = O(L^{-1/2})$.

Before the first failure, the probability of entering stage j is $(1 + o(1))p_j^0$ uniformly, since the relative error in every conditional failure probability is $o(1)$ and $Jq = \gamma$. The expected ideal reward up to and including the first failure is therefore $(1 + o(1))\sqrt{(1-\varepsilon)T_0}$. If the retiling loss probability summed over the branch is $O(m^{-9})$, as arranged below, all three types of artificial truncation have probability $o(1)$. Cauchy–Schwarz and the reward second moment show that they change the mean reward by $o(L^{1/4})$. Converting from reward to the sum of the spatial block coefficients gives

$$\mathbb{E}_\nu \sum_{\text{retained steps}} \sqrt{r_j} \ge (1 - o(1))\sqrt{n}\,\sqrt{(1-\varepsilon)T_0}, \tag{4.8}$$

because $r_j^{-1/2} = \sqrt{n\lambda_j}$.

For the terminal cost, the conditional ν-mean of $\log(1+\beta)$ equals $ds/2 + O(ds \sup|\beta|)$ by expanding $(1+\beta)\log(1+\beta)$ under the ordinary law. The conditional variance is $O(ds)$, and the centered jumps are $O(m^{-c_0})$. In stage j the total actual clock, including previous stages, is bounded by $CS_j \log J$. The centered sum of the log increments up through each such stage, stopped at its caps, is at most $m^{-\kappa/4} S_j$ in absolute value except with probability $\exp(-m^{c_1})$ for some $c_1 > 0$, by the bounded-increment martingale tail inequality. To obtain it, apply the same conditional exponential bound as for bounded independent sums. If d_i are the centered martingale increments, then for $|t| \max_i \|d_i\|_\infty \le 1$, $\exp(t\sum_{i\le j} d_i - Ct^2 \sum_{i\le j} \mathbb{E}[d_i^2 \mid \mathcal{F}_{i-1}])$ is a supermartingale. Stop it at the first threshold crossing and optimize t, using both signs. Here the predictable quadratic variation bound

is $CS_j \log J$, and $m^{-\kappa/2} S_j / \log J \geq m^{\kappa/2} / \log J$. Union over j preserves an exponential bound, and the accumulated drift errors are $o(S_j)$. Also the clock of earlier stages is $O(S_j L^{-4} \log J) = o(S_j)$. Thus on one joint good event, in stage j one has $\log Y = s_j/2 + o(S_j)$, with s_j the clock spent in that stage.

A completed success has relative ratio at most D_j. On failure in stage j the actual ratio satisfies $|M| \leq (1 + o(1))b_j$, including all prior overshoots. Apply Cauchy–Schwarz with (4.5) and $p_j = (1 + o(1))q$. Conditional on ideal entry, the expectation on actual failure and on the joint good clock event is bounded by

$$(1 + o(1))b_j \sqrt{(1 + o(1))q} \sqrt{(1 + o(1))qS_j/2 + o(qS_j)} = (1 + o(1))qB/\sqrt{2}.$$

This uses a bound on the integrand on the joint good event only, so no conditional estimate on the previous value of $\log Y$ at a rare entry history is needed. The terminal cost on time, geometry or clock stops is $O(B\sqrt{\log J})$ on this same event, with expected contribution $o(1)$ from the above bounds. On completion of all stages, $|M| \leq \sum_j D_j = o(b_J/\sqrt{\log J})$ gives another $o(1)$ cost. On the bad clock event the deterministic bound $|\log Y| \leq C\sqrt{n \log R\, S_J \log J}$ follows by Cauchy–Schwarz on the increments. Its contribution is negligible because the event has exponentially small probability. Summing the entry probabilities times the failure costs gives

$$\mathbb{E}_\nu[|M| \sqrt{(\log Y)_+}] \leq (1 + o(1)) \frac{B}{\sqrt{2}} (1 - e^{-\gamma}). \tag{4.9}$$

The same bounds give $\mathbb{E}_\nu[M^2 \log(2 + Y|M|/\eta_*)] = O(L)$ for any fixed $\eta_* > 0$. Indeed on the joint good event the logarithm is bounded by $C(1 + S_j \log J)$ in the stopping stage and $|M| \leq Cb_j$. On its complement the preceding deterministic bound again suffices.

4.4 Realizing the tilted process by a test potential

Begin with $u_0(x) = x_2 - 1/2$ on Ω. At a step, a branch cell is an old affine triangle (or the initial square), and its gradient and stage parameters are constant. Choose A by

$$A^{-\mathsf{T}} = \begin{pmatrix} a_j^{-1/4} & M a_j^{1/4} \\ 0 & a_j^{1/4} \end{pmatrix}.$$

This has determinant one. Both A and A^{-1} have norm bounded by fixed powers of m since a_j and a_j^{-1} do by (4.2) and $|x| \leq b_j$, and $|M| \leq 2b_1$. The same polynomial bound holds for the change between successive charts. In the new chart retain all grid squares inside the old cell with side the old intrinsic leg times a dyadic factor comparable to m^{-D_0}, for a sufficiently large fixed D_0. The old intrinsic leg of a triangle is the side of the square in which it was one of the four reference triangles. An uncovered point lies within a constant times the new side of the old boundary. Because the aspect ratio in the new chart is polynomial, the ratio of boundary-strip area to old area is $O(m^{-10})$ for large enough D_0. Freeze the potential on the uncovered remainder. By the pathwise tilt identity the lost tilted probability over $O(m)$ steps is $O(m^{-9})$.

On each new chart square $Q = x_Q + dAS$, write p_Q for the pullback of a reference function p by $x \mapsto d^{-1}A^{-1}(x - x_Q)$, extended by zero. Reveal the finitely many Gaussian coefficients $d^{-1}W(g_{Q,k})$, where $g_{Q,k} = (g_k)_Q$. Build U on S from these reference coefficients and add $dY\sqrt{r_j}U_Q$ to the potential. Its physical gradient increment is the specified matrix applied to ξ, by the chain rule. Its mean and integrated gradient are zero. Its support is strictly inside Q. The potential therefore

remains globally continuous piecewise affine, hence Lipschitz for each finite mesh, and its vertical derivative stays positive since $\sup|\beta| = o(1)$.

Each reveal is predictable. For a new square, every previous direction is either zero there or affine there. This follows from the nested placement inside previous child triangles, since all the old g's are piecewise affine on those triangles. The new directions have zero affine moments. They are orthogonal to all previous revealed directions and to the constant function. The directions in disjoint new squares are orthogonal as well. To justify conditional freshness, at each step orthonormalize the new finite family given the old data. Conditional on that data, the remaining white noise is isonormal on the orthogonal complement of the revealed space. Conditioning further on the new independent coordinates leaves it isonormal on the new complement. This induction starts with the ordinary white noise on the square and establishes the claimed reference law and all the conditional estimates for the chosen charts.

A microstep costs at most $\log R + D_0 \log m + O(1)$ of intrinsic log-depth. Choose n to be the largest integer whose cumulative bound is at most $(1-\epsilon_m)m$, with $\epsilon_m = (\log m)^{-1/2}$. Then $n \sim m/\log R$ and every used chart leg is at least $d_{\min} = h^{1-\epsilon_m}$ up to an absolute factor. This is enough to pair against $P_h W^0$. Conditional on the old data, the normalized scalar pairing of a block differs from its unprojected expected pairing only by $o(1)$ before division by $\sqrt{v}$. Indeed, the sum of the unclipped diagonal covariance differences is bounded by

$$C \sum_k \lambda_k^{-1}(h\|A^{-1}\|/d)\|\nabla f_k\|_2 \le C(h\|A^{-1}\|/d)K_{\max} = o(1).$$

The estimate uses cell Poincaré for P_h and the zero extension of f_k, which is supported inside Q. For a predictable adaptive test function $P_h f_{Q,k}$, write its noise pairing as its conditional old component plus the remaining centered Gaussian pairing. The old component makes zero contribution in expectation since the fresh input coefficients are odd and the clipping event is even. The remaining component has variance no greater than $\|P_h f_{Q,k}\|_2^2 \le Cd^2$. Fourth moments and the clipping probability then bound the normalized pairing loss by $O(R^{-5}\log R)$ before division by $\sqrt{v}$. Thus the conditional expected contribution to $W^0(P_h u)$ from Q is

$$(1+o(1))d^2 Y \sqrt{r_j}\sqrt{\log R/\pi},$$

uniformly in the old history. The factors $d^2 Y$ are exactly the probabilities of the squares under the tilted spatial branch law. The initial affine potential has expected pairing zero. Sum over the retained steps and use (4.8) to obtain

$$\mathbb{E}W^0(P_h u) \;\ge\; (1-o(1))\sqrt{m/\pi}\,\sqrt{(1-\varepsilon)T_0}.$$

4.5 Duality and optimization

Let $D_*(u) = 1 + \eta_* + \int_\Omega \eta_* \Theta^*(|\partial_1 u|/\eta_*)$ for fixed $\eta_* > 0$, where Θ^* is the convex conjugate of $\Theta(t) = e^{t^2} - 1$. If a current has gauge at most r, apply the weak divergence identity against u, use Fenchel in the first coordinate, and bound the second coordinate by $r\int_\Omega |\partial_2 u| = r$. This gives $|W^0(P_h u)| \le r D_*(u)$. To use a Lipschitz test in the weak identity, extend it by reflection, smooth, and pass to the limit in the integrals against the integrable current using bounded gradients and almost everywhere convergence. Infimizing over the currents gives the same bound with $r = \mathcal{F}(P_h W^0)$.

For $v \ge 0$, the maximizing $t \ge 0$ in $\Theta^*(v)$ satisfies $v = 2te^{t^2}$, so $\Theta^*(v) \le v\sqrt{\log(2+v)}$. Indeed the bound on t is immediate for $t \le \sqrt{\log 2}$, and otherwise $2t > 1$ implies $t^2 \le \log v$. Since

$X = YM$ and $|M| \le 2b_1 = o(1)$, the integral in D_* is at most the conditional spatial Y-average of $|M|\sqrt{(\log Y)_+} + o(1)$. For example, $\log(2 + Y|M|/\eta_*) \le (\log Y)_+ + \log(2 + 2b_1/\eta_*)$. Equation (4.9) and Jensen under the probability density Y now give

$$\mathbb{E}D_*(u) \le 1 + \eta_* + (1 + o(1))\frac{B}{\sqrt{2}}(1 - e^{-\gamma}), \qquad \mathbb{E}D_*(u)^2 = O(L).$$

The second bound uses the terminal estimate for $\mathbb{E}_\nu[M^2 \log(2 + Y|M|/\eta_*)]$ above.

The grid cut estimate (3.1) and affine interpolation bound the flux gauge of the histogram with cell values g_v/h by $C\|g\|_2$ for mean-zero g. Here we use $\|J_1\|_{\psi_2} \le \|J_1\|_\infty/\sqrt{\log 2}$. The gauge is subadditive by the triangle inequality, so it is uniformly Lipschitz in the Gaussian cell coefficients. Gaussian concentration gives $\mathrm{Var}(\mathcal{F}(P_h W^0)) = O(1)$. Cauchy–Schwarz therefore bounds the covariance of the gauge with D_* by $O(\sqrt{L})$, negligible in the pairing bound. With (4.7) and $L \sim \log\log m$, we obtain

$$\liminf_{h\downarrow 0} \frac{\mathbb{E}\mathcal{F}(P_h W^0)}{\sqrt{m}(\log\log m)^{1/4}} \;\ge\; \frac{1}{\sqrt{\pi}} \frac{\sqrt{(1-\varepsilon)(B/\sqrt{2})(1 - e^{-2\gamma})}}{1 + \eta_* + (B/\sqrt{2})(1 - e^{-\gamma})}.$$

Set $B = \sqrt{2}/(1 - e^{-\gamma})$, then let $\varepsilon, \eta_*, \gamma$ tend to zero after the mesh limit. The right side tends to $1/\sqrt{2\pi}$. The variance bound also gives the lower bound with this constant in probability.

5 The sharp upper bound

The upper construction uses a thick Gaussian refinement step in a determinant-one chart. Its principal singularity gives the variance coefficient $1/(4\pi)$. The following phase and stability estimates retain that coefficient through the predictable chart changes.

5.1 A sharp continuous phase

Let $M = \log(1/h)$ and, throughout this subsection, write $a = \log\log M + 10\log\log\log M \sim \log\log M$, $a_0 = a^{1/10}$, $u_0 = a^{-1/10}$, $L_0 = a^{1/4}$, and $K = a^4$. Choose a smooth $\alpha(u)$, constant a_0 for $u \le a^2$, increasing to a over a window of length $O(a)$ in $\log u$, and constant afterwards. Such a ramp is obtained by integrating a smooth bump on an interval of length Ca, with C a large fixed constant. Its first derivative in $\log u$ is in $[0, 1/8]$, and the higher derivatives used below are bounded. Put

$$r = e^{-\alpha(u)}\sqrt{\frac{u}{1+u}}(1 + u/M)^{-1/4}, \qquad I(u) = K\int_0^u r(v)\frac{dv}{v}.$$

For $u \le 1$, $r(u) \le e^{-a_0}\sqrt{u}$. From 1 to the end of the ramp the logarithmic interval has length $O(a)$ and $r \le e^{-a_0}$. Beyond that point, integration up to M costs at most $e^{-a}\log M$, and the factor $(1 + u/M)^{-1/4}$ bounds the remaining integral by Ce^{-a}. Thus $I(\infty) = o(1)$. Let

$$h(t, y) = \frac{L(t)^2 - y^2}{L_0^2}, \qquad L(t) = L_0 + 2(1 + \delta_M)\sqrt{a}\, t,$$

where $\delta_M \to 0$ will dominate errors in the derivative bounds. Use the implicit phase

$$-h(t, y) = I(u) - r(u)(e^{\alpha(u)x^2/u} - 1)$$

on the strip $h + I(\infty) > 0$, setting $u = 0$ on the central line where no positive solution exists. The right side is strictly increasing in u. Write $s = x/\sqrt{u}$, $z = \alpha s^2$, $J = e^z$, $\beta = \partial_{\log u} \log r$, $\gamma = \partial_{\log u} \log \alpha$, $d = 1 - \gamma$, and

$$p = 2\alpha s r J, \quad q = 2\alpha r J(1 + 2z), \quad Q = r[K - \beta(J - 1) + dzJ], \quad D = q(Q + sQ_s/2 - Q_{\log u}) - Q_s^2$$

where $Q_{\log u}$ is at fixed s. We have $\beta \in [-3/8, 1/2]$, $|\beta'| \lesssim 1$, and $|\gamma| + |\gamma'| \lesssim 1/a_0$. Hence $Q \asymp r(K + zJ)$. The bounds below hold uniformly in u, z.

$$\frac{q}{Q} \le 4\alpha(1 + o(1)), \qquad \frac{p^2}{Q} \le 4\alpha(1 + o(1))rJ, \qquad 0 \le \frac{D}{Q^2} \lesssim \alpha, \qquad \frac{u\widehat{H}_{xx}}{\widehat{H}_Y} \lesssim \alpha r J(1 + \log K).$$

Here $\widehat{H}(x, Y)$ is the implicit level for the graph in (x, Y), so $H(t, x, y) = \widehat{H}(x, -h(t, y))$. For the first two bounds, split at $z = (\log K)/2$. Below this value the numerator relative to the rK term is negligible. Above it, use $dz - \beta \ge (1 - o(1))z$. To check the other two, put $k = K + \beta$, $v = dz - \beta$, $w = v + d$, $A = (1 - \beta)k - \beta'$, and $E = v(1 + v) + d^2 z - d'z + \beta'$. Then

$$Q = r(k + Jv), \quad Q_s = wp, \quad Q + sQ_s/2 - Q_{\log u} = r(A + JE)$$

and

$$\frac{D}{2\alpha r^2 J} = (1 + 2z)A + J\big[(1 + 2z)E - 2zw^2\big].$$

The bracket is a quadratic in z with uniformly bounded coefficients, leading coefficient $2d - d^2 - 2d' \ge 1/2$. For bounded z its negative part is absorbed by $A \asymp K$; for large z it is positive. Together with $q > 0$, this makes the implicit Hessian positive semidefinite. Indeed, in the coordinates (x, u) the quadratic form for $D^2\widehat{H}/\widehat{H}_Y$ has diagonal entries q/u and $(Q + sQ_s/2 - Q_{\log u})/u^2$, off-diagonal entry $-Q_s/u^{3/2}$, and determinant D/u^3. Also

$$\frac{u\widehat{H}_{xx}}{2\alpha r J \widehat{H}_Y} = (1 + 2z) - \frac{4zJw}{k + Jv} + \frac{2zJ(A + JE)}{(k + Jv)^2}.$$

For $z \le 1 + \log K$ use $Q \asymp r(K + zJ)$. For larger z, replace $k + Jv$ by Jv at error $O(K/J)$, and use $1 - 2w/v + E/v^2 = O(1/z)$. This gives the stated bound.

Let $B_H = D^2 H + \nabla H \nabla H^{\mathsf{T}}$. For positive levels the graph identities give

$$\frac{\det B_H}{\widehat{H}_Y^2} = h_y^2 \left(\frac{q}{Q} + \frac{D}{uQ^2}\right) + (-h_{yy}) \left(\frac{\widehat{H}_{xx}}{\widehat{H}_Y} + \frac{p^2}{Q}\right), \qquad -H_t = h_t \widehat{H}_Y.$$

For $u \ge a^2$, the displayed derivative bounds therefore give

$$\frac{\sqrt{\det B_H}}{\widehat{H}_Y} \le 2\sqrt{a}(1 + o(1))\sqrt{h_y^2 - h_{yy}(h + I + r)} \le 4\sqrt{a}(1 + o(1))L/L_0^2.$$

Indeed $rJ = h + I + r$, and the last bound uses $|y|^2 \le L^2 + L_0^2 I(\infty)$. For $u_0 \le u \le a^2$ the same inequality holds with a strict $o(\sqrt{a})$ coefficient, since $\alpha = a_0$ and $a_0(1 + \log K)/u_0 = o(a)$.

Flatten H by replacing it with $\chi(H)$, where $\chi' = 0$ on $[0, u_0/2]$, $\chi' = 1$ on $[u_0, \infty)$, and χ' increases smoothly between these regions with

$$\chi'' \le C u_0^{-1} \chi'(1 + \log^2(1/\chi')).$$

Such a cutoff follows by starting its derivative like $e^{-1/v}$ in the rescaled transition coordinate. In the transition with $\chi' \ge e^{-4a_0}$, the determinant of $B_{\chi(H)} = \chi' D^2 H + (\chi'' + \chi'^2)\nabla H \nabla H^{\mathsf{T}}$, divided by $(\chi' \widehat{H}_Y)^2$, gains a factor $\chi''/\chi' + \chi'$ only on the gradient rank-one term. This follows by the same graph formula with q/Q and p^2/Q multiplied by that factor. Its square root is still bounded by $o(\sqrt{a})\sqrt{h_y^2 - h_{yy}(h + I + r)}$, since $a_0^3/u_0 = o(a)$. In the remaining transition, the two coefficients χ' and χ'' are at most $e^{-4a_0}\,\mathrm{poly}(a)$ and $\widehat{H}_Y = u/Q \le e^{a_0}\,\mathrm{poly}(a)$. The determinant root is then $o(1)$ absolutely. Set $F = \chi(H) + a^{-1}(T - t)$, with $T = 1 + o(1)$. Choose $\delta_M \downarrow 0$ to dominate the relative errors in the high-level bounds, for example a sufficiently large multiple of $(\log a)^{-1/4}$. The low-level bound $e^{-3a_0}\,\mathrm{poly}(a) = o(a^{-1})$ is absorbed by the time-only term. For $B_0 = D^2 F + \nabla F \nabla F^{\mathsf{T}}$, use the regularized matrix

$$B = B_0 + \frac{\epsilon_{\mathrm{reg}}^2}{1 + \operatorname{tr} B_0} I, \qquad \epsilon_{\mathrm{reg}} = (\log M)^{-1}.$$

The increase in determinant root is $O(\epsilon_{\mathrm{reg}})$. The bounds just proved give $-F_t \ge (1 + \omega_M)\sqrt{\det B}$ for a slowly vanishing $\omega_M > 0$ after choosing the wall-speed margin as above. They are uniform over $0 \le t \le T$.

On $0 \le H \le M$ the same level equation gives

$$x^2 \le u + \frac{u}{\alpha(u)} \left[O(\log a) + \tfrac{1}{2} \log(1 + 1/u) + \tfrac{1}{4} \log(1 + u/M) \right] \le (1 + o(1))u + o(1).$$

Here $\alpha \ge a_0$, $u \log(1+1/u) \le 1$, and $\chi(u) = u + O(u_0)$. Hence $x^2 \le (1+o(1))\chi(H) + o(1)$ on the active states and their retained one-step overshoots. The strip has vertical radius $L_0 + 2(1+o(1))\sqrt{a}\,T + o(1)$. These estimates, together with the determinant bound, are the phase inputs used below.

5.2 Discrete stability of the sharp continuous phase

We prove the pointwise conditional estimate that allows the sharp continuous phase to run along the adaptive spatial construction. All vectors in this subsection are in the unbalanced coordinates of the upper bound. The preliminary determinant-one balancing map takes the initial domain to a rectangle tiled by squares in these coordinates. The initial means on these squares are handled by a separate current. The field entering the phase starts at zero.

Use $M = m$ in Section 5.1. Denote the deterministic time horizon by $T_{\mathrm{end}} = 1 + o(1)$ and the normalized variance allowance per split by $\delta \asymp (\log m)^2/m$. Write $u = H(t, z)$, $v = \nabla_z F(t, z)$, and

$$B_0 = D_z^2 F + vv^{\mathsf{T}}, \qquad \eta = \frac{\epsilon_{\mathrm{reg}}^2}{1 + \operatorname{tr} B_0}, \qquad B = B_0 + \eta I, \qquad K = \sqrt{\det B}, \qquad C = KB^{-1}.$$

The selector C is positive definite with determinant one. There is a slowly vanishing $\omega = \omega_m > 0$ for which $-F_t \ge (1 + \omega)K$, and $F(0, 0) = o(1)$. We use the deterministic level cutoff $U_* = m/(\log m)^A$, with A a large fixed constant.

Bounds for the selector. In the region $u \le 2U_*$ use widths $b_x = \sqrt{1 + u}$ and $b_y = 1$. In statements about derivatives one can smoothly replace u by zero for $u \le u_0/4$, and leave it as u for $u \ge u_0/2$. This does not affect any of the estimates for F, which is spatially flat for $u \le u_0/2$. For some fixed power P of $\log m$, independent of A,

$$\frac{1 + u}{P} \le K \le P(1 + u), \qquad C_{ii} \le P b_i^2, \qquad |b_i b_j F_{ij}| \le P(1 + u). \tag{5.1}$$

The last bound holds also after applying ∂_t or one of $b_k\partial_k$ to F_{ij} before multiplying by b_ib_j. These operators applied to $\log(1+u)$ and $\log K$ are bounded by P. Here and below P can be increased by a fixed power.

To verify these bounds, let $Y=-h(t,y)$, $H_Y=u/Q$, and $v_H=\nabla H$. Use the profile quantities p,q,Q,D,s of Section 5.1, with the profile constant there denoted by $K_r=a^4$. Differentiating the inverse graph gives the exact positive semidefinite decomposition

$$D^2H=H_Y\left[\frac{q}{u}\left(e_x-\frac{Q_s}{q\sqrt{u}}v_H\right)\left(e_x-\frac{Q_s}{q\sqrt{u}}v_H\right)^{\mathsf T}+\frac{D}{qu^2}v_Hv_H^{\mathsf T}+Y_{yy}e_ye_y^{\mathsf T}\right]. \tag{5.2}$$

Indeed, for the function $Y=G(u,x)$ defining the graph, $G_{xx}=-q/u$, $G_{xu}=Q_s/u^{3/2}$, and $G_{uu}=(Q_{\log u}-sQ_s/2-Q)/u^2$. Implicit differentiation and completing the square give (5.2). Also $H_x=\sqrt{u}\,p/Q$, $H_y=uY_y/Q$ and $H_t=uY_t/Q$.

For $u\ge u_0/4$, $r^{-1}\le(\log m)^{O(1)}$ because $\alpha\le a$ and $e^a=(\log m)(\log\log m)^{10}$. On the strip $rJ=h+I+r=O(\sqrt{a})$. Thus J, s, z, Q^{-1}, q^{-1}, Y_y, Y_{yy}, and all the numerators in the graph formulas are bounded by powers of $\log m$. The formula for D gives

$$D\gtrsim\alpha r^2J\big[(1+z)K_r+(1+z^2)J\big].$$

For bounded z the term with K_r absorbs the bracket's bounded negative part. For sufficiently large z the bracket has its uniform positive quadratic lower bound. Using the three first derivatives of H just computed shows that $\log u$ and $s=x/\sqrt{u}$ have derivatives bounded by powers of $\log m$ in the scaled directions and in time. The ramp of α has bounded derivatives through order four in $\log u$, so the same bounds apply to the derivatives of q,Q,D and to their logarithmic derivatives. These assertions follow by differentiating their explicit formulas. The displayed positive lower bounds for q,Q,D ensure no larger reciprocal appears.

The three vector factors of (5.2), with the square roots of their scalar coefficients included, have coordinates bounded by $P(1,\sqrt{1+u})$, together with their scaled derivatives and their time derivatives. For example the first scalar coefficient is q/Q, and $Q_s/(q\sqrt{u})$ and its derivatives have size at most $P/\sqrt{1+u}$. The second scalar coefficient is $D/(quQ)$ and has square root at most $P/\sqrt{1+u}$. The third factor is $\sqrt{uY_{yy}/Q}\,e_y$. Meanwhile v_H and its corresponding derivatives are bounded coordinatewise by $P(\sqrt{1+u},1+u)$. The cutoff χ can be chosen so the square roots of χ' and χ'' have smooth, polynomially bounded derivatives in $1/u_0$. For example, choose χ'' as a normalized positive flat exponential bump on the transition interval, then integrate twice with zero initial data. This proves the Hessian estimates for $F=\chi(H)+\epsilon_0(T_{\mathrm{end}}-t)$.

For $u\ge u_0$ the determinant identity in the continuous phase gives $\det B_0=(u/Q)^2\mathcal{D}$, where $\mathcal{D}$ and its derivatives in the stated directions are polynomially bounded in $\log m$. Also $\mathcal{D}\ge P^{-1}$. When $|y|\ge L(t)/2$, use h_y^2q/Q. Otherwise h is bounded below by an absolute positive constant, so $rJ=h+I+r$ is bounded below and $z=\log J$ is bounded away from zero. Then use $Y_{yy}p^2/Q$. The change from adding ηI in the determinant is $\epsilon_{\mathrm{reg}}^2\operatorname{tr}B_0/(1+\operatorname{tr}B_0)+\eta^2$. Its scaled and time derivatives are polynomially bounded, since $\operatorname{tr}B_0\gtrsim(1+u)/P$ and its derivatives are at most $P(1+u)^2$. In the cutoff layer all entries and derivatives are polynomially bounded and $\det B\ge\epsilon_{\mathrm{reg}}^4/(1+\operatorname{tr}B_0)^2$. This gives the bounds for $K,\log K$. The cofactor formula for C and the coordinate bounds on the Hessian factors and gradient give $C_{ii}\le Pb_i^2$.

For a quantitative estimate when the selector moves, let S_C have rows given by the three Hessian factors in (5.2) multiplied by $\sqrt{\chi'}$, the two vectors $\sqrt{\chi''}v_H$ and $\chi'v_H$, and the two identity rows multiplied by $\sqrt{\eta}$, all divided by $\sqrt{K}$. Thus $S_C^{\mathsf T}S_C=C^{-1}$. For any fixed determinant-one metric G with $G_{ii}\le P_1b_i^2$, where P_1 is a fixed power of $\log m$, the derivatives of $S_CG^{1/2}$ in the same scaled

and time directions are at most $P_2\sqrt{1+u}$. For each row this follows by the coordinate bounds and $\|e_i^{\mathsf{T}} G^{1/2}\| = \sqrt{G_{ii}}$. For the regularization rows use

$$|\partial\sqrt{\eta}| \le \frac{\epsilon_{\mathrm{reg}}|\partial \operatorname{tr} B_0|}{2(1+\operatorname{tr} B_0)^{3/2}} \le P\sqrt{1+u}$$

outside the cutoff layer and polynomial bounds inside it.

The induction on active cells. At generation j, $t = j\delta$ and the accumulated regular field is Z_j. The metric G of a parent is the product of its chart map with its transpose. Its determinant is one. All root cells have $G = I$, $Z_0 = 0$ and selector $C = I$. Select an interior representative x_Q of each active cell measurably from the old history. All pointwise statements are on cell interiors, with cell boundaries omitted from spatial integrals. Freeze terminal cells and cells whose representative has $H(t, Z_j(x_Q)) \ge U_*/2$. At each other parent set $C_Q = C(t, Z_j(x_Q))$, diagonalize $G^{-1/2} C_Q G^{-1/2}$, clip its two reciprocal eigenvalues to $[K_0^{-1}, K_0]$, and conjugate back by $G^{1/2}$. The result G_+ satisfies

$$\det G_+ = 1, \qquad K_0^{-1} G \preceq G_+ \preceq K_0 G, \qquad G_+ \preceq K_0^{-1} G + C_Q. \tag{5.3}$$

The last inequality follows from $\mathrm{clip}(\lambda) \le K_0^{-1} + \lambda$ for each eigenvalue.

We use the following properties of the two local splits in the assembly. At fixed x, the uncut increment is Gaussian with covariance at most δG_+. Its coarse and fine summands are independent conditional on the old history, and each is multiplied by an even scalar in $[0,1]$ depending only on its own Gaussian data. There is a nonnegative retained envelope $E_j(x)$ with $E_j(x) \le CE_0$, $E_0 = \sqrt{m}/(\log m)^B$, and with conditional tail Ce^{-cv^2} at v. The retained increment has chart magnitude at most $C\sqrt{\delta}E_j(x)$ and a Hölder bound with a fixed exponent and the same coefficient on the child containing x, with distances divided by its chart leg. For the coarse increment this follows from its bound on the containing base cell. For the fine increment it follows from its bound on the fine child. A rejected fine base cell contributes zero throughout that cell. The assembly and the local estimates verify these assertions before either reveal at this step. The child contraction in chart diameter, combined with the fixed relative distortion in (5.3), can be taken as small as desired for descendants that split again.

Define the past envelope sum $T_j(x)$ by $T_0 = 0$ and $T_{j+1} = \rho T_j + E_j^2$, with fixed $0 < \rho < 1$. Along an active history assume the relative width changes at each past step are at most $1 + o(1)$, and maintain $G_{ii} \le P_1 b_i(t, Z_j(x))^2$ at every point in the cell. For x, x' in one old cell the ancestor Hölder bounds imply

$$\sum_i \frac{|Z_j(x)_i - Z_j(x')_i|}{b_i(t, Z_j(x))} \le P_2\sqrt{\delta}\,\sqrt{T_j(x)}. \tag{5.4}$$

To see this, the distance between x and x' in the chart of generation $i < j$, divided by the leg of their common child at that generation, is at most $C\vartheta^{j-i-1}$ for a fixed small ϑ. Apply the ancestor Hölder estimate there. On converting that chart increment to coordinate k, the factor is at most $\sqrt{P_1} b_k(t_i, Z_i(x))$. Its ratio to the current width is bounded by $C(1+c)^{j-i}$ for a small fixed c. Choose ρ larger than the square of the resulting geometric decay ratio and use Cauchy–Schwarz. This gives (5.4), and $T_j \lesssim E_0^2$.

For clarity, the deterministic induction starts with $T_0 = 0$ and all widths equal to one. Immediately after any split the old states and their new values, including the time advance, have level at most $2U_*$, as follows. Use the scaled derivative bounds above along the short segments in state and time. A state displacement at most $CP_2\sqrt{\delta}E_0$ in the widths changes $1+u$ and K by a relative $o(\omega)$, and a time change δ has the same property. This follows by a continuation argument on the

first segment where the width ratio lies in $[1/2, 2]$, because the logarithmic derivatives are bounded by P. The segments remain in the strip. At positive levels its margin in the variable Y is

$$I(\infty) - I(u) + r(J-1) \;\geq\; I(\infty) - I(u) \;\gtrsim K_r r(u),$$

by integrating on $[u, 2u]$. This is at least an inverse power of $\log m$ for $u_0/4 \leq u \leq 2U_*$. At lower levels the margin is bounded below by $I(\infty) - I(u_0/4)$. The same estimates and the explicit polynomial bounds for Y_t, Y_y handle the short change of time and the vertical coordinate. Take B sufficiently large for these estimates.

In particular, if the representative has level below $U_*/2$, (5.4), applied with the representative as the first point, shows that all old states in that cell have level below U_*. In this application the derivative continuation begins at the representative, so it requires no pointwise upper-level assumption beyond the induction for the preceding step. The pointwise widths at the representative and any point of the old cell are relatively $1 + o(1)$. By (5.3) and (5.1), $G_{+,ii} \leq (1 - c_1) P_1 b_i^2$, for some fixed $c_1 > 0$ and P_1 sufficiently larger than P. The fresh retained increment consequently has normalized width displacement at most $C\sqrt{P_1 \delta} E_j(x)$. The same continuation estimate keeps the new level below $2U_*$ and closes the metric cap and width induction. Geometric terminal cells reached by a split already satisfy these estimates and are frozen thereafter. If a representative triggers the high-state stop, (5.4) and the same continuation estimate show that the entire cell has level at least $U_*/3$, for large m.

Lag of the implemented metric. For positive determinant-one matrices set $\mathcal{R}(G, C) = \operatorname{tr}(GC^{-1}) - 2$. If the larger relative eigenvalue is e^d, then $\sqrt{\mathcal{R}(G, C)} = 2 \sinh(d/2)$. For clipping at a fixed target, d becomes $(d - \log K_0)_+$, and $\sinh((d - \log K_0)/2) \leq K_0^{-1/2} \sinh(d/2)$ when $d \geq \log K_0$. Thus clipping contracts the square-root lag by $K_0^{-1/2}$ without an additive term. Also

$$\sqrt{\mathcal{R}(G, C)} = \sigma - \sigma^{-1}, \qquad \sigma = \|S_C G^{1/2}\|_{\mathrm{op}} \geq 1.$$

This expression is 2-Lipschitz in the rectangular matrix factor, since operator norm is 1-Lipschitz and $s - s^{-1}$ has derivative at most two on $[1, \infty)$.

Write $\mathcal{R} = \mathcal{R}(G, C(t, Z_j(x)))$, $\mathcal{R}_+ = \mathcal{R}(G_+, C(t, Z_j(x)))$, and $\mathcal{R}' = \mathcal{R}(G_+, C(t + \delta, Z_{j+1}(x)))$. Let $\zeta = P_3 \sqrt{\delta(1 + U_*)}$ for a sufficiently large fixed power P_3 of $\log m$. The rectangular factor derivative bound and (5.4) show that the square-root lag changes by at most $C\zeta\sqrt{T_j}$ when the selector is moved to or from the representative, with either G or G_+ held fixed. The state and time advances change it by $C\zeta(E_j + 1)$ with G_+ held fixed. Use the strict clipping contraction between the two representative comparisons, and square with a fixed small slack. For constants $q_1, q_2 < 1$,

$$\mathcal{R}_+ \leq q_1 \mathcal{R} + C\zeta^2 T_j, \qquad \mathcal{R}' \leq q_2 \mathcal{R} + C\zeta^2 (T_j + E_j^2 + 1). \tag{5.5}$$

Take A large enough that ζ tends to zero faster than any inverse-logarithmic power required below. For $S_j = \mathcal{R} + \zeta T_j$, the second inequality and the discount recursion give

$$S_{j+1} \leq q_3 S_j + C\zeta(1 + E_j^2), \qquad q_3 < 1.$$

Since $K' = K(1 + o(\omega))$ at the end of the step, we obtain

$$K' S_{j+1} - K S_j \leq -cK S_j + C\zeta K (1 + E_j^2). \tag{5.6}$$

One-step exponential moment. At an old point with state $z = Z_j(x)$, put $\xi = Z_{j+1}(x) - z$. By the derivative bounds and the uniform clipped displacement cap, Taylor's formula gives

$$F(t, z+\xi) - F(t, z) \le v \cdot \xi + \tfrac{1}{2}\xi^{\mathsf{T}} D_+ \xi, \qquad D_+ = D^2 F(t, z) + e_m K \operatorname{diag}(b_i^{-2}).$$

Here $e_m P_1 = o(\omega)$, by taking B large. Indeed the Hessian changes along the segment by at most $P(1+u)$ times the relative width displacement in weighted matrix norm, and $1+u \le PK$. The added diagonal dominates that change. The phase is convex, so D_+ is positive semidefinite. Advancing time at the new state gives

$$F(t+\delta, z+\xi) - F(t, z+\xi) \le -(1+\omega/2)\delta K,$$

by the time inequality and comparability of K.

For any convex test Φ and an even scalar attenuation $a \in [0,1]$ of a symmetric Gaussian vector g,

$$\mathbb{E}\Phi(ag) = \tfrac{1}{2}\mathbb{E}[\Phi(ag) + \Phi(-ag)] \le \tfrac{1}{2}\mathbb{E}[\Phi(g) + \Phi(-g)] = \mathbb{E}\Phi(g).$$

In fact the pair $ag, -ag$ consists of opposite convex combinations of $g, -g$. With two independent Gaussian contrasts and separate attenuations, apply this twice, conditioning each time on the other summand. Thus the exponential of the linear plus positive quadratic Taylor bound has moment at most that of the uncut Gaussian sum.

Here is the quantitative Gaussian estimate. For a centered Gaussian vector of covariance $\Sigma \preceq \delta G_+$, completion of the square gives, for p near one,

$$\frac{1}{p}\log \mathbb{E} e^{p(v\cdot g + g^{\mathsf{T}} D_+ g/2)} = -\frac{1}{2p}\log\det(I - pA) + \frac{p}{2} v^{\mathsf{T}} \Sigma^{1/2}(I-pA)^{-1}\Sigma^{1/2} v, \qquad A = \Sigma^{1/2} D_+ \Sigma^{1/2}.$$

The formula is valid also for singular Σ by using a standard Gaussian and the symmetric square root. The scaled Hessian and metric bounds give $\|A\| \le \delta P(1+2U_*) = o(\omega)$. Expanding the determinant and bounding the inverse show that for $p = 1 + o(\omega)$ this expression is at most

$$(1+o(\omega))\frac{\delta}{2}\operatorname{tr}[G_+(D_+ + vv^{\mathsf{T}})] \le (1+o(\omega))\delta K(1 + \mathcal{R}_+/2) + o(\omega\delta K).$$

The last inequality uses $B_0 \preceq B = KC^{-1}$ and the weighted bound $e_m P_1 = o(\omega)$.

Now use the exponent $\Xi_j = F(t, Z_j) + \kappa\delta K S_j$. Apply Hölder with $p = 1+\tau$ and its conjugate $p' = p/(p-1)$ to the Taylor moment and the positive memory term in (5.6). Choose $\zeta/\tau = o(1)$ and $\tau = o(\omega)$. Since E_j has a uniform conditional Gaussian tail, $\log \mathbb{E}[e^{\theta(1+E_j^2)} \mid \mathcal{F}_j] \le C\theta$ for $\theta \ge 0$ sufficiently small, uniformly in the old history $\mathcal{F}_j$. Take $\theta = Cp'\kappa\delta\zeta K$, which is small. The conjugate Hölder factor then costs $O(\kappa\delta\zeta K) = o(\omega\delta K)$ in the logarithm. The first bound in (5.5) gives $\mathcal{R}_+ \le CS_j$. Choose fixed κ large enough that the negative memory drift pays the lag part of the moment. The remaining bound is at most

$$-(1+\omega/2)\delta K + (1+o(\omega))\delta K + o(\omega\delta K) \le 0.$$

Consequently $\mathbb{E}[e^{\Xi_{j+1}(x)} \mid \mathcal{F}_j] \le e^{\Xi_j(x)}$ at every active point. The exponent is frozen in full on every stopped cell.

Conditional integration over space gives a supermartingale of total modular at least one, since the area is one and all terms in Ξ_j are nonnegative. At zero the selector and the metric are both the identity, and there is no memory. The initial modular is therefore $e^{F(0,0)} = 1 + o(1)$. It follows by Markov's inequality on its nonnegative excess that the terminal modular is $1 + o_{\mathbb{P}}(1)$. On the same event, each high-stop cell carries exponent at least $U_*/3 - O(u_0)$ throughout. Their union has area at most e^{-cU_*}. Every stored state has $H \le 2U_*$ and lies inside the moving strip, so the horizontal level estimate and the vertical strip bound from Section 5.1 apply to the entire retained regular field.

5.3 Routing the rare stopped regions

Let $U_* = m/(\log m)^A$, and condition on a main exploration history for which the high-state stopped parents cover a set $\mathcal{H}$ of area $p \le e^{-cU_*}$. The case $p = 0$ requires no additional exploration. The unrevealed sources in the stopped parents are independent white noise bridges by predictable orthogonal revelation. The old metrics, in physical balanced coordinates, satisfy

$$G_{Q,xx} \le (1 + U_*)\,\mathrm{polylog}(m), \qquad G_{Q,yy} \le \mathrm{polylog}(m).$$

All estimates below are conditional on that history. We build a current for the additional contrasts and leave only terminal bridges for the final mesh correction.

Putting the roots in one chart. Put $a_b = 1+\log(1+\log(m/U_*))$ and set $A_b = \mathrm{diag}(a_b^{-1/4}, a_b^{1/4})$ and $G_b = A_b A_b^{\mathsf{T}}$. On the stopped parents, successively clip the old metric towards G_b with relative eigenvalue ratio K_0. Concretely, clip the eigenvalues of $G^{-1/2} G_b G^{-1/2}$ to $[K_0^{-1}, K_0]$ and conjugate back by $G^{1/2}$. The clipped metric is determinant one, lies between $K_0^{-1}G$ and $K_0 G$, and satisfies $G_+ \preceq K_0^{-1}G + G_b$, as follows from the same scalar inequalities on the eigenvalues. The logarithm of the larger eigenvalue of $G^{-1/2} G_b G^{-1/2}$ decreases by $\log K_0$, down to zero. Consequently every nonterminal descendant reaches G_b in $O(\log m)$ splits. At each step use the ordinary subdivision and ray current of Lemmas 2 and 3, with the common minimum leg s. At the geometric cutoff retain the cell for the terminal correction. Use even local clipping at $\sqrt{m}/P$, where P is a sufficiently large fixed power of $\log m$. The regular current has chart supremum $C\sqrt{m}/P$ per step. Iteration of the matrix inequality gives along each branch

$$\sum_j \sqrt{G_{j,ii}} \le C\sqrt{G_{0,ii}} + C(\log m)\sqrt{G_{b,ii}}.$$

After division by $\sqrt{m}$, the regular steering current therefore has vertical supremum $o(1)$ and horizontal supremum at most $(1 + U_*)^{1/2}\,\mathrm{polylog}(m)/P$ on $\mathcal{H}$. For a field of supremum M_0 on a set of area p, the quadratic Luxemburg norm is at most $M_0/\sqrt{\log(1 + 1/p)}$. This follows by substituting that scale in its modular. The horizontal steering norm is $o(1)$. For the removed pieces use Lemma 8 with the current divided by $\sqrt{m}$, threshold $\sqrt{m}/P$ and box enlargement $P^2 \log m$. The removed currents have both normalized gauges $o(1)$ with probability $1 - o(1)$. These corrections use the revealed contrasts. Conditional on the steering exploration, the remaining bridges are fresh. In $A_b^{-1}\Omega$, the nonterminal roots have metric I, and we start their regular backup currents at zero.

A phase for the remaining bridges. Fix $0 < \alpha < 1/10$, to be chosen with a clipping exponent $\alpha/2 < \eta < 1/10$. Define

$$r(u) = \frac{\sqrt{u}}{(1+u)^{1/2+\alpha}} + \frac{1}{1+\log(m/U_*)}\,\frac{(u/U_*)^\alpha}{1+(u/U_*)^\alpha}(1+u/m)^{-\alpha} + \frac{(u/m)^\alpha}{(1+(u/m)^\alpha)^2}, \qquad I(u) = \int_0^u r(v)\,\frac{dv}{v}.$$

The first and third summands have bounded integrals in du/u. For the middle summand integrate separately over $(0, U_*)$, $[U_*, m]$, and (m, ∞), using the power decay on the two tails and the prefactor on the middle interval. Thus $I(\infty) \le C_\alpha$. The elasticity $\beta = d\log r / d\log u$ lies in $[-\alpha, 1/2]$ and has uniformly bounded derivative in $\log u$. Each summand has these bounds, and the derivative of their

weighted mean adds the variance of the individual elasticities. For any fixed C_1, the third summand yields

$$r(u) \gtrsim_{C_1} m^{-\alpha}, \qquad \frac{1+u}{mr(u)} \lesssim_{C_1} 1, \qquad 1/2 \le u \le C_1 m. \tag{5.7}$$

Use $Y(y) = \sqrt{1+y^2} - 1$, $c(t) = c_0 + \Lambda t$, where c_0 is a sufficiently large fixed constant. On the strip $Y(y) < c(t) + I(\infty)$ define $u = H(t, x, y)$ by the graph

$$Y(y) = c(t) + I(u) - r(u)(e^{x^2/u} - 1).$$

Set $H = 0$ at $x = 0$, $Y(y) \le c(t)$. Write $z = x^2/u$, $J = e^z$, $Z = 1 + z$, $Q = r[1 - \beta(J-1) + zJ]$, $D_0 = Q/(rJ)$. The graph derivative in u is Q/u, and $Q \asymp rJZ > 0$, using $0 \le J - 1 \le zJ$ and $\beta \le 1/2$. Its endpoint limits give the asserted unique level. On these levels $rJ \le C$ for $\Lambda t \le 1$ and $Q \lesssim Z$. For $u \ge 1$ the first summand gives $z \le C_\alpha \log(2+u)$, hence $Z \lesssim u$. For $1/2 \le u \le 1$ it bounds r below and z above. Take a smooth nondecreasing cutoff χ that vanishes below $1/2$ and has derivative one for $u \ge 1$, and set $H_0 = 1 + \chi(H)$.

For $u \ge 1$ put

$$b_x^2 = uZ/Q, \qquad b_y^2 = Q/Z, \qquad K = 1 + u/Q,$$
$$D = C_D \operatorname{diag}(1/Z, uZ/Q^2), \qquad v = \nabla H_0,$$
$$C = \sqrt{\det(D + vv^{\mathsf{T}})}(D + vv^{\mathsf{T}})^{-1}.$$

Extend the widths, K and the two diagonal entries of D by smooth positive interpolation to constants below $1/2$. On the flat region take $D = C_D I$. The derivative bounds below have constant versions on the compact transition, since u, z, Q^{-1} are bounded there. The flattened phase and these coefficients are smooth, including across the zero-level set.

Implicit differentiation on $u \ge 1$ gives

$$H_x = 2x/D_0, \qquad H_y = uY'/Q, \qquad H_t = -\Lambda u/Q.$$

From $D_0 = z - \beta + (1+\beta)e^{-z}$ one obtains

$$|z_x| \lesssim |x|/(uZ), \qquad z_y = -zY'/Q, \qquad z_t = \Lambda z/Q.$$

The first derivatives of D_0 in z, β are bounded. The derivatives of $\log Q$ in z and in $\log u$ at fixed z are bounded by the explicit formula for Q. Differentiating the two gradient formulas, using $\partial_i \beta = \beta' H_i/u$ and $|Y'|, |Y''| \le 1$, gives

$$|H_{xx}| \lesssim Z^{-1}, \qquad |H_{xy}| \lesssim \sqrt{u}/Q, \qquad |H_{yy}| \lesssim uZ/Q^2.$$

For the last bound, $|Q_y/Q| \lesssim Z/Q$ and $Q \lesssim Z$ handle respectively the terms with Q_y and Y''. It follows that $D \succeq D^2 H_0$ for large C_D and $|b_i b_j \partial_{ij} H_0| \lesssim K$. For $u \ge 1$, $v^{\mathsf{T}} D^{-1} v \asymp u$. The upper bound follows from the gradient formulas. When z is bounded away from zero, $ZH_x^2 \gtrsim u$. For small z, the graph and the choice of c_0 give $Y(y) \ge c_0 - O(1)$ and $|Y'| \gtrsim 1$, so $Q^2 H_y^2/(uZ) \gtrsim u$. With $b_x^2 b_y^2 = u$ and $Z \lesssim u$, this gives $\sqrt{\det(D + vv^{\mathsf{T}})} \asymp K \asymp u/Q$. In particular for $u \ge 1$

$$C^{-1} = \xi\big(C_D \operatorname{diag}(b_i^{-2}) + ww^{\mathsf{T}}\big), \qquad w = \sqrt{Q/u}\, v, \qquad \xi \asymp 1, \qquad |b_i w_i| \lesssim \sqrt{u}.$$

The cofactor formula and $\det C = 1$ give $C_{ii} \lesssim b_i^2$, also in the extended region.

For a determinant-one G with $G_{ii} \le M_0 b_i^2$ write $\mathcal{L}(G) = \sqrt{\operatorname{tr}(GC^{-1})}$. We record the derivative estimates used for the metric lag. On $u \ge 1$,

$$\begin{aligned} \mathcal{L}^2 &\lesssim_{M_0} 1+u, \qquad v^{\mathsf T} G v \lesssim K\mathcal{L}^2, \\ |b_j \partial_j \mathcal{L}| &\lesssim_{M_0} \sqrt{(1+u)/r}, \qquad |\partial_t \mathcal{L}| \lesssim_{M_0} \Lambda\sqrt{1+u}/r. \end{aligned} \tag{5.8}$$

In the flattened region use a fixed positive r in these bounds. To verify the derivatives, the preceding formulas bound $b_j\partial_j$ of $\log u, z, \log Q, \log b_i, \log K$ by $C/\sqrt{r}$, and their time derivatives by $C\Lambda/r$. For instance $b_x|z_x| \lesssim \sqrt{z/(QZ)}$, and $b_y|z_y| \lesssim z/\sqrt{QZ}$. The Hessian and the explicit gradients then bound $|b_i b_j \partial_j w_i|$ by $C\sqrt{u/r}$ and $|b_i \partial_t w_i|$ by $C\Lambda\sqrt{u}/r$. The same scaled derivative bounds as for $\log Q$ hold for ξ, by $\xi^{-2} = C_D^2/u + C_D \sum_i (b_i w_i)^2/u$. Finally $\mathcal{L}$ is the Euclidean norm of the vector with entries $\sqrt{\xi C_D G_{ii}}/b_i$ and the two entries of $\sqrt{\xi} G^{1/2} w$. This proves the derivative bounds without division by a small component. The first two inequalities in (5.8) follow from the selector expression, using $D + vv^{\mathsf T} \succeq vv^{\mathsf T}$ for the second.

Discrete evolution and modular. For the backup in its chart take $\delta = \varepsilon^2/m$ and time $t = j\delta$, with $\varepsilon > 0$ a sufficiently small fixed constant. There are at most $C_* m$ stages by the geometric shrink, so fix $T_* = C_* m \delta$. Update each active parent metric by relative clipping towards C evaluated at its representative. Take the ordinary split with minimum leg s and the ray inverse of the fresh contrast. Multiply its chart current by $\sqrt{\delta}$, applying even local clipping at $E_0 = m^{1/2-\eta}$ before the phase update. Write E for the resulting local envelope without $\sqrt{\delta}$. The clipping gives $E \le CE_0$ and $\mathbb{P}(E > t \mid \text{old data}) \le Ce^{-ct^2}$. The spatial state U is the sum of the retained increments, with the chart maps applied to the increments. Stop and freeze the entire integrand below when a cell reaches the geometric cutoff. We also freeze globally at the first stage whose modular exceeds a fixed multiple of p.

Here are the geometric and analytic details of the induction on active cells. Maintain the cap $G_{ii} \le M_0 b_i^2$ at each point of a parent, for a sufficiently large fixed M_0, and keep the widths between successive states within an arbitrarily small fixed relative factor. For x, x' in the same old cell, the ancestor Hölder bounds give

$$|U_i(x) - U_i(x')| \le C\sqrt{\delta}\, b_i(t, U(x))\sqrt{\mathcal{T}(x)}. \tag{$*$}$$

Here $\mathcal{T}$ is a discounted sum of squared past envelopes, with update $\mathcal{T}' = \rho\mathcal{T} + E^2$ for some $\rho < 1$. To see the estimate, compare the two points in the chart of each ancestor. The fixed chart-distortion bound per step and the sufficiently small child shrink make their relative distance decay geometrically with the number of intervening levels. The widths can grow by at most the chosen small fixed factor per level. The resulting geometrically weighted sum of envelopes is bounded by $C\sqrt{\mathcal{T}}$ by Cauchy–Schwarz, with ρ strictly larger than the squared geometric ratio. The retained envelopes give $\mathcal{T} \lesssim E_0^2$. The first root has $U = 0$, $G = I$, and $\mathcal{T} = 0$.

Before a modular stop, each active cell has a point with $H \le C_1 m$. Its area is at least cs^2, and the modular dominates the integral of e^{H_0}, so this follows from the modular cap $Cp \le C$. Use $(*)$ from this witness to the representative and every point in the parent. The scaled derivative bounds above give relative changes $o(1)$ in $1 + H, b_i, K$ on these segments. Here the spatial parameter is $\sqrt{\delta}E_0/\sqrt{r} = O(m^{\alpha/2-\eta}) = o(1)$ by (5.7). For rigor, impose factor-two bounds on the coefficients and $1 + H$ along the segment, apply the derivative bounds up to their first possible failure using the floor up to $2C_1 m + 2$, then continue. The wall margin in the Y variable is

$$I(\infty) - I(u) + r(u)(J-1) \gtrsim rJ \asymp b_y^2.$$

For the first term integrate on $[u, 2u]$, where bounded elasticity keeps r comparable. The displacement in the vertical direction is $O(\sqrt{\delta} b_y E_0) = o(b_y^2)$ since $b_y \gtrsim \sqrt{r}$. Thus the segment stays in the strip, with a fixed positive margin used in the flattened region. The time step costs $O(\Lambda\delta/r) = o(1)$ in relative changes and widens the strip. This proves the estimates at all old active points without presupposing a pointwise upper bound on H.

Representative comparison and $G_+ \preceq K_0^{-1} G + C$ now give a strict margin in the diagonal cap for the new metric. The fresh displacement is $O(\sqrt{\delta} b_i E)$, so the same continuation argument closes the diagonal cap at the new state and bounds the relative width changes. It also bounds the one-step overshoot of H by Cm. All chart distortions and physical cell axes have polynomial bounds. This follows from the formulas for b_i, (5.7) and the determinant-one caps, including the low-level extension. Thus the terminal cells have physical diameter $h\,\mathrm{poly}(m)$ by the geometric cutoff. These bounds include the $O(\log m)$ steering generations.

To check the lag update, first evaluate both metrics at the representative. If $e^d \geq 1$ is the larger relative eigenvalue, then the squared lag is $e^d + e^{-d}$. Clipping decreases d by $\log K_0$, down to zero. Consequently the lag norm contracts by $K_0^{-1/2}$ up to a bounded additive constant. Changes from the representative and on the fresh spatial and time segments have, by (5.8), total bound

$$C\sqrt{\delta(1+u)/r}(\sqrt{\mathcal{T}} + E) + C\Lambda\delta\sqrt{1+u}/r \leq C\varepsilon(\sqrt{\mathcal{T}} + E) + C\Lambda\varepsilon^2.$$

For the last inequality use (5.7) and its flat-region version. Take $S = \mathcal{L}^2 + \mathcal{T}$. Squaring with slack in the clipping contraction and using the discount update gives $S' \leq q_0 S + C(1 + E^2)$ with $q_0 < 1$, for small ε with $\Lambda\varepsilon^2 \leq 1$. The relative change of K then gives

$$K'S' - KS \leq -cKS + CK(1 + E^2). \tag{$**$}$$

Use the exponent

$$\Xi = H_0(t, U) + C_0(T_* - t) + \kappa\delta KS.$$

For the fresh step move first in space, then in time. The Taylor remainder in space is at most $C\delta K E^2$ by the Hessian bound and comparability along the segment. For the predictable linear tilt the uncut conditional Gaussian variance is at most $C\delta K\mathcal{L}^2$, by (5.8), the coarse covariance bound and the relative bound between G_+ and G. Even scalar attenuation gives the same upper bound for the linear exponential moment by averaging the two Gaussian signs. In time the derivative of H_0 equals $-\Lambda u/Q$ above the transition, and is nonpositive in the transition. The scalar clock covers the transition and flat region for $C_0 \geq C\Lambda$. By comparability the combined time contribution is at most $-c\Lambda\delta K$. Together with $(**)$, conditional Cauchy–Schwarz bounds the log moment of $\Xi' - \Xi$ by

$$-c\Lambda\delta K - c\kappa\delta KS + C\delta K\mathcal{L}^2 + C_\kappa\delta K.$$

Indeed, the envelope tail gives $\log \mathbb{E}[\exp(aE^2) \mid \text{old data}] \leq Ca$ for all sufficiently small fixed $a \geq 0$, and $\delta K \lesssim \varepsilon^2$ by the floor bound. Choose κ to pay the term with $\mathcal{L}^2$ and then Λ, C_0 to pay the bounded term. Choose ε last, so the envelope moment bound holds and $\Lambda T_* \leq 1$. All constants in the phase estimates were uniform for this time interval. The resulting log moment is nonpositive.

Spatial integration with the prescribed freezing proves that $\int_{\mathcal{H}} e^{\Xi}$ is a nonnegative supermartingale. Initially $H_0 = 1$, the scalar clock is bounded, and $K, \mathcal{L}$ are bounded at $U = 0, G = I$. The initial integral is therefore at most Cp, with an absolute bound after shrinking ε. Choose the modular stop at a sufficiently large fixed multiple of p. The maximal inequality gives conditional probability bounded below of completing without this stop. On this event $H \leq Cm$ by the area witness and overshoot estimate, $\int_{\mathcal{H}} e^H \leq Cp$, and the normalized backup vertical component is bounded by the strip width C.

Cost and exact divergence. On $1/2 \le u \le Cm$ the middle profile term yields

$$\log(C/r(u)) \le C' + \log(1 + \log(m/U_*)) + \alpha \log_+(U_*/u).$$

Indeed $(1 + u/m)^{-\alpha}$ is bounded below here and $v/(1+v) \ge \frac{1}{2}\min(v, 1)$ for $v = (u/U_*)^\alpha$. Using $rJ \le C$ and $u \log_+(U_*/u) \le U_*/e$, we get

$$x^2 \lesssim a_b(1 + u) + U_*.$$

This bound holds also for $u < 1/2$, because $r(u) \gtrsim \sqrt{u}$ there bounds $u \log(C/r(u))$. At scale $C_2\sqrt{a_b}$, with C_2 sufficiently large, the horizontal exponential integral on $\mathcal{H}$ is at most $Cp \exp(CU_*/(C_2^2 a_b)) = o(1)$, by $\int_{\mathcal{H}} e^H \le Cp$ and $p \le e^{-cU_*}$. Outside $\mathcal{H}$ the backup current is zero. Thus its normalized horizontal ψ_2 cost is $O(\sqrt{a_b})$.

Lemma 8 applied in the backup chart to the removed pieces, with $E_0 = m^{1/2-\eta}$ and box enlargement $Cm^{2\eta}$, gives normalized hard vertical cost $O(1)$ and horizontal ψ_2 cost $O(m^{5\eta-1/2}) = o(1)$. These estimates and the steering repairs hold on events of conditional probability $1 - o(1)$, uniformly before the modular stop, so intersecting with them still leaves a fixed positive conditional success probability. The sum of the retained and repaired currents has divergence equal to the sum of all revealed contrasts, because each repair replaces exactly the divergence removed by clipping. Undo the factor $\sqrt{\delta}$ and push the backup forward by A_b. Its physical costs in both directions are $O(\sqrt{m}\, a_b^{1/4})$, while the steering costs are $o(\sqrt{m})$. Since $a_b = O(1 + \log\log\log m) = o(\log\log m)$, these are lower order at the sharp scale.

5.4 Assembly and exact repair

Write $m = \log(1/h)$, $a = \log\log m$, and $\lambda = m/(4\pi)$. The target is

$$\mathcal{F}(P_h W^0) \le (1 + o(1))\sqrt{m/(2\pi)}\, a^{1/4}$$

with probability bounded away from zero. Here $\mathcal{F}$ is the continuum gauge with zero normal flux. Uniform Lipschitz concentration then gives the same asymptotic upper bound in expectation and in probability. Indeed affine interpolation of a discrete cut-capacity solution proves $\mathcal{F}(h^{-1}g) \le C\|g\|_2$ for a centered fine-grid cell vector g, so the Lipschitz constant with respect to the standard Gaussian cell coefficients is uniform.

Set $L \sim (\log m)^2$, $R = e^L$ an integer, and $s = hm^{C_s}$. Choose a determinant-one balancing map $A_0 = \mathrm{diag}(b, b^{-1})$ with b^2 an integer and $b = (1 + o(1))\sqrt{2}\, a^{1/4}$. In $A_0^{-1}\Omega$ tile the area-one rectangle by b^2 squares of side $1/b$. Reveal their coarse means. In physical coordinates their density is constant on horizontal strips, and its vertical primitive has supremum $O_{\mathbb{P}}(1)$. In fact the successive strip masses before centering are independent Gaussians of variance $1/b^2$. If S_j is their partial sum, the primitive at strip boundaries is $S_j - (j/b^2)S_{b^2}$. The L^2 martingale maximal inequality bounds its supremum in L^2. The regular evolution starts at zero, independently of these coarse means.

At a stage, before revealing new contrasts, stop a parent on the high-state criterion $H(t, U(x_Q)) \ge U_*/2$ with $U_* = m/(\log m)^A$. The within-parent phase comparison of the discrete stability estimates justifies this representative test. On each other parent clip its old metric towards the sharp selector with the fixed relative ratio K_0. In the new chart take an ordinary fixed-cap Whitney split with minimum side Rs. On every nonterminal base square S refine further by the uniform $R \times R$ grid. Both partitions are chosen before new reveals. The coarse child-average contrast and the centered fine contrasts inside each base square are orthogonal Gaussian projections, with independent fine contrasts in different base squares. Use the ray inverse of Section 2 for the coarse contrast and the inverse with radial principal part for each fine contrast. All fine children are squares in the current

metric chart. Every child of a coarse split receives the coarse update, including the terminal base cells. These terminal cells receive no fine contrast and freeze after the coarse update. A parent too small for a thick split freezes directly and its unrevealed bridge goes to the terminal repair.

Clip the coarse increment by the even envelope rule in Lemma 8 at $\sqrt{m}/P$, with P a sufficiently large power of $\log m$. For the fine contrast clip a whole base square if any fine-child envelope, with amplitude and oscillation divided by $\sqrt{L}$, exceeds $\sqrt{m}/P$. On an accepted base square retain the entire fine current. It has zero normal flux on that square, so rejecting it removes exactly its fine source. Each fine rejection has conditional probability at most $\exp(-cm/P^2)$ by the Gaussian envelope bounds and the union over R^2 children. These events are conditionally independent over base squares, including after the coarse reveal. The coarse and fine attenuation rules depend separately on their own innovations and are even in those innovations.

The chart covariance of the uncut sum is bounded above by $(L/(4\pi)+O(\sqrt{L}))I$ by the isotropic-principal lemma and the $O(1)$ coarse covariance. Divide the physical chart increment by $\sqrt{\lambda}$ in the phase evolution, so the step clock is $\delta = (L+O(\sqrt{L}))/m$. Every nonterminal branch shrinks by a factor $O(R^{-1})$ in intrinsic leg per step, since the relative metric distortion is fixed. The depth is at most $(m+O(\log m))/(L-O(1))$. The sharp phase can therefore run over a deterministic horizon $T = 1+o(1)$. Its discrete stability gives terminal horizontal ψ_2 norm at most $1+o(1)$ and vertical supremum at most $(2+o(1))\sqrt{a}$ in normalized unbalanced coordinates. The former follows from $x^2 \le (1+o(1))H+o(1)$ on the active levels and their one-step overshoots, and the terminal modular $1+o_{\mathbb{P}}(1)$. The latter follows from the moving strip, whose final radius is $L_0+2(1+o(1))\sqrt{a}\,T+o(1)$. These bounds include the stored regular values on the high-stop cells. Their union has area at most $\exp(-cU_*)$. The balancing map and the factor $\sqrt{\lambda}$ give exactly $(1+o(1))\sqrt{m/(2\pi)}a^{1/4}$ in both physical gauges.

The removed fine sources. Use physical determinant-one repair maps $D_k = \mathrm{diag}(k, k^{-1})$, with a sufficiently large polylogarithmic k to be fixed below. For each candidate base square, in $D_k^{-1}\Omega$ choose a predictably enlarged axis square containing its transformed support. Its side is at most a polynomial in m times the intrinsic base leg. If it cannot be fit in the domain, the base leg is inverse-polynomial and the conditional rejection union bound rules out rejection of any such square with probability $1-o(1)$. For a fitting candidate subdivide the repair square dyadically down to width g, comparable to its fine leg times a fixed inverse polynomial in m. The logarithm of the number of mesh cells per side is $L+O(\log m) = o(\sqrt{m})$.

For every candidate fine contrast f, even before testing its rejection, the Gaussian vector of its projected repair-cell masses divided by g has covariance at most the identity. The coordinate tests are disjoint cell indicators divided by g, composed with the determinant-one map and projected onto the fine contrast space in its original base square. Orthogonal projection is a contraction. The sum of the masses is zero in the repair square. The capacity estimate of Lemma 5 and Gaussian concentration give common repair load $O(\sqrt{m})$ with conditional failure probability e^{-Cm} for arbitrarily large fixed C. Union over all attempted candidates. The residual $(I-P_g)f$ is mean zero on each repair cell and costs at most $Cg\|f\|_\infty$ by coordinate primitives. The Gaussian union bound gives $\|f\|_\infty \le C\sqrt{m}$ divided by the fine leg, so this cost is $O(\sqrt{m})$. Thus on the joint good event, the exact repair per rejected square has physical horizontal supremum $O(k\sqrt{m})$ and vertical supremum $O(\sqrt{m}/k)$ by pushing forward D_k.

At any point and stage, polynomially many candidate repair boxes cover it, by disjoint area packing in dyadic intrinsic size classes and polynomial distortion. Conditional independence of fine rejections gives the exponential overlap bound by expanding the one-stage counting moment at tilt cm/P^2. Iterating over the stages and union over a fine net after padding boxes gives a uniform

overlap $O(P^2)$. The expected sum of rejected box areas is polynomial times e^{-cm/P^2}, so their union has area at most $e^{-c'm/P^2}$ with high probability. Choose $k/P^2 \to \infty$ polylogarithmically. The vertical repair is $o(\sqrt{m})$. The horizontal supremum is polylogarithmic times $\sqrt{m}$ on this sparse support, hence its global ψ_2 norm is polylogarithmic and is also $o(\sqrt{m})$. Lemma 8, applied to the unscaled coarse current with threshold $\sqrt{m}/P$ and repair stretch $k \gg P^2$ polynomial in $\log m$, gives vertical cost $o(\sqrt{m})$ and horizontal Luxemburg cost $o(\sqrt{m})$ for the removed coarse sources. Treat the two types separately, so no joint coarse-fine rejection estimate is needed.

Completion. Use the shifted-tail backup on the stopped union, conditional on the full main history. It succeeds at fixed positive conditional probability, with both physical budgets $o(\sqrt{m}a^{1/4})$. Its initial steering increments and their sparse repair have smaller order as well. Steering and backup use the same minimum side s as the main exploration. The terminal leaves of the combined exploration have physical diameter at most $hR\,\mathrm{poly}(m)$. Before the last residual projection, the revealed density satisfies $\|f_{\mathrm{leaf}}\|_\infty \le Cm^{3/2}/s$ with high probability. Each contrast has density standard deviation at most C/s on its new cells, all of which have area $\gtrsim s^2$. A Gaussian union bound over $O(m)$ generations with at most Cs^{-2} cells per generation gives this estimate by summing the contrasts along the branches. The initial root-square means satisfy the same bound. Thus the deterministic correction $(P_h - I)f_{\mathrm{leaf}}$ costs $O(\sqrt{m})$ by coordinate primitives on the final grid. For the remaining bridges choose an h-aligned dyadic grouping grid of side D comparable to the terminal diameter bound and larger than h plus that bound. Assign each leaf to the grouping square containing a representative of the leaf. Enclose the square and its immediate neighbors in a square of side $4D$ with sides on the h-grid. At the boundary slide that square inside Ω. Each enlarged square is within $O(D)$ of the original one, so their overlap is bounded. They contain the final mesh projections of every assigned bridge. Each projected bridge has zero mass. Conditional on the exploration, the normalized mesh masses of the sum of the assigned bridges have covariance at most the identity, by orthogonal projection onto their unrevealed Gaussian subspaces. Lemma 5 thus gives mean capacity $O(\log(D/h)) = O(L + \log m) = o(\sqrt{m})$ in each box. Uniform Gaussian concentration and a union bound give total repair cost $O(\sqrt{m})$. All conditional bridges are valid since the geometry, stopping, and discarded increments have used only already revealed masses. Sum the coarse means, main regular current, exact removed-source repairs, backup and terminal currents. Their divergence is exactly $P_h W^0$.

The total current has both gauges at most $(1 + o(1))\sqrt{m/(2\pi)}\, a^{1/4}$ on an event of probability bounded below. By the uniform Gaussian concentration of the gauge, its mean is at most this bound plus $O(1)$, and the same bound with a relative $o(1)$ holds in probability. Together with the lower bound of Section 4, this proves Theorem 2.

6 An asymptotically exact Gaussian reduction

We first compare the point cost with the flux gauge of an empirical histogram. The upper inequality needs only local count estimates. Once we couple that histogram to white noise, the Gaussian upper bound gives the order estimate required for the reverse inequality.

Theorem 3. *Put $m = \log N$, $s_N = \sqrt{m/N}$ and $\rho_N = (\log\log m)^{1/4}$. For ℓ dyadic with $\ell \asymp N^{-1/2}m^{10}$, the endpoint cost and white noise can be coupled so that*

$$M_N = \sqrt{2/N}\,\mathcal{F}(P_\ell W^0) + O_{\mathbb{P}}(s_N). \tag{6.1}$$

In particular, if $\ell = 2^{-j}$, then

$$\frac{M_N}{s_N \rho_N} - \frac{\mathbb{E}\mathcal{F}(P_{2^{-j}} W^0)}{\sqrt{j \log 2}\,(\log\log j)^{1/4}} \xrightarrow{\mathbb{P}} 0. \tag{6.2}$$

The same statement holds after exchanging the two directions.

The proof uses the conditional count estimate of Lemma 6 and the Gaussian upper bound proved above. All costs use the convention in Theorem 1.

6.1 From point measures to a smooth flux gauge

Let μ_N^+, μ_N^- be the two empirical probability measures and choose a dyadic $h \asymp C_h s_N$ with a sufficiently large fixed C_h. Let S_ℓ smooth by reflection on the square. Explicitly, extend the source to the period-two torus by even reflection in each coordinate, then convolve with a nonnegative, reflection-symmetric kernel K_ℓ of integral one, supported in a ball of radius $C\ell$. Take $\|K_\ell\|_\infty \le C\ell^{-2}$ and $\|\nabla K_\ell\|_\infty \le C\ell^{-3}$. For vector currents reflect the normal component oddly and the tangential component evenly. This extension commutes with divergence, and the smoothed current has zero normal trace on the square.

Uniformization. Each sample couples to its P_ℓ histogram with coordinatewise maximum displacement $O(s_N)$ with probability tending to one. First spread each point uniformly in its h-cell. The P_h histogram is bounded below by a positive absolute constant on the uniform occupancy event. Within each ℓ-square, compare the fine counts to their mean over the square. Conditional on its total, these are multinomial counts with uniform cell labels. Lemma 6 gives a signed flow with both face loads $O(Nh^2)$, uniformly over the ℓ-squares with probability tending to one. The estimate applies since the boxes have polynomially many h-cells per side, their conditional cell means are $O(Nh^2)$, and the conditional failure bound has arbitrarily large exponential rate in m after increasing the load bound. The lower bound on $P_h\mu_N^\pm$ also bounds $P_\ell\mu_N^\pm$ below. The face interpolation gives a zero-normal flux of supremum $O(h)$ for the difference of the two histograms. Linear interpolation of the densities and velocity equal to flux divided by density then give the desired coupling. For a measurable flux this velocity argument follows by first smoothing with the same reflection convention at a scale tending to zero and taking a weak limit of couplings.

Chernoff's inequality on the coarse grid also gives

$$\|P_\ell \mu_N^\pm - 1\|_\infty = O(s_N/\ell)$$

with probability tending to one. The histogram p couples to $S_\ell p$ at maximum displacement $O(s_N)$. The evenly extended density has translation flux

$$J(z) = \int K_\ell(v)\, v \int_0^1 (p(z - tv) - 1)\, dt\, dv$$

with divergence $p - S_\ell p$ and supremum at most $C\ell\|p - 1\|_\infty$. The parity of J gives the boundary condition, and the linear-density velocity again gives the coupling. Finally reflection smoothing is contractive for the coordinatewise maximum transport distance. Add the same random displacement and fold both positions back into the square with the coordinatewise period-two reflection map, which is 1-Lipschitz in each coordinate. By gluing, each sample couples to its S_ℓ density at maximum displacement $O(s_N)$.

For use in the other direction, take a coupling whose starting marginal is one of the samples and whose maximum displacement is $D = o(\ell)$. Spread its straight paths with the reflected kernel.

The time-t occupation density r_t then differs uniformly from the smoothed starting density by $O(D/\ell)$. Indeed, only starting atoms within distance $C\ell$ of the evaluation point or its reflections can contribute. These have total mass $O(\ell^2)$ by the coarse occupancy cap. The derivative bound for the kernel now gives the estimate. Apply this also to the point-to-P_ℓ coupling with $D = O(s_N)$. It follows that

$$\|S_\ell \mu_N^{\pm} - 1\|_\infty = O(s_N/\ell). \tag{6.3}$$

Sharp upper transfer. For densities p, q each bounded below by $1-\epsilon$, a zero-normal flux of divergence $p-q$ and gauge r gives a coupling of gauge at most $r/(1-\epsilon)$. For the velocity J/ρ_t along $\rho_t = (1-t)p + tq$, convexity gives

$$\int \left(e^{((1-\epsilon)\Delta_1/r)^2} - 1\right) d\pi \le \int_0^1 \int \rho_t \left(e^{((1-\epsilon)J_1/(r\rho_t))^2} - 1\right) dx\, dt \le (1-\epsilon) \int \left(e^{(J_1/r)^2} - 1\right) dx.$$

The hard bound follows by integrating the absolute vertical velocity on each path. Smooth approximation as above proves this for general feasible fluxes. Use (6.3) and glue to the point measures with additive cost $O(s_N)$. The resulting fractional point coupling has the hard displacement cap as a support constraint. Multiply its matrix by N, giving row and column sums one. Any row subset has at least as many positive-support neighbors as rows by the two marginal conditions, so Hall's theorem gives a supported permutation. Subtract the smallest entry on that permutation, rescale the remaining common marginal, and repeat. Each subtraction either finishes the decomposition or removes a positive entry. The horizontal modular is linear in the coupling, so one of the supported permutations has modular at most that of the coupling. It follows that

$$M_N \le (1 + O_{\mathbb{P}}(s_N/\ell))\mathcal{F}(S_\ell(\mu_N^+ - \mu_N^-)) + O_{\mathbb{P}}(s_N).$$

The smoothed flux gauge may be replaced by the histogram one with additive error $O_{\mathbb{P}}(s_N)$. For each color spread the paths of its point-to-P_ℓ coupling. Its smoothed occupation density is uniformly bounded, and the resulting flux has both suprema $O(s_N)$. It connects $S_\ell \mu_N^{\pm}$ to $S_\ell P_\ell \mu_N^{\pm}$. The translation flux then connects the latter to $P_\ell \mu_N^{\pm}$ at the same cost. By subadditivity and symmetry of $\mathcal{F}$, the two empirical flux gauges differ by $O_{\mathbb{P}}(s_N)$.

6.2 Sharp coupling of the histogram counts

Poissonize the coarse cell counts of each color, so they are independent Poisson variables of mean $\lambda = N\ell^2$. Each Poisson count can be coupled by its quantile to a standard normal Z so that

$$|P - \lambda - \sqrt{\lambda} Z| \le C m^{3/2} \qquad \text{on } |Z| \le 4\sqrt{m}. \tag{6.4}$$

Here only $\lambda \asymp m^{20}$ is needed. For a proof put $R = 4\sqrt{m}$. Stirling's formula, expanding the Poisson rate function at λ, compares each mass at $|k-\lambda| \le 8R\sqrt{\lambda}$ to its normal cell mass with relative error $O((1+R^3)/\sqrt{\lambda})$. The normal cell uses endpoints $(k - \lambda \pm 1/2)/\sqrt{\lambda}$. The Poisson and normal tails beyond this range are negligible relative to the relative-error tolerance times either retained tail at a threshold in $[-2R, 2R]$, by Chernoff's inequality. Summing the mass estimate gives the same relative error for both distribution tails in that threshold range. The lower normal tail has logarithmic derivative bounded away from zero on $(-\infty, 1]$, and the upper tail has the analogous bound on $[-1, \infty)$. Hence thresholds $Z \pm C(1+R^3)/\sqrt{\lambda}$, with the rounding error included, bracket the Poisson quantile. This proves (6.4).

Couple the cells independently and truncate each coupling error to zero on $|Z| > 4\sqrt{m}$. With probability tending to one this changes none of the cells. Center the errors by removing the spatial

mean of the histogram. For any dyadic box of side $d \geq \ell$, the sum of errors centered by their expectations is $O(m^2 d/\ell)$ uniformly over the boxes, with probability tending to one, by Hoeffding and a union bound. Spatial centering costs only a constant factor in this estimate. The histogram residual divided by N therefore has a flux with both suprema $O(m^3/(N\ell))$. In detail, telescope the averages over the dyadic levels. For a mean-zero child-average difference in a box of side d, row primitives after subtracting row means, followed by a vertical primitive of those row means, have suprema at most Cd times the maximum child density. Each level costs $O(m^2/(N\ell))$ and there are $O(m)$ levels. This residual is $o(s_N)$.

To restore fixed sample size, condition on the Poisson total K and put the Poissonized points in independent uniform order, then append independent uniform points as needed. Conditional on K this sequence is iid uniform, so taking its first N points gives the required law. The difference has $|K - N| \leq \sqrt{N}m$ extra or removed uniform labels with probability tending to one. If there are k such labels, the centered count in each dyadic box of side d is $O(\sqrt{kd^2m} + m)$ uniformly by Bernstein. The same dyadic primitives for this centered correction cost at most

$$Cm\sqrt{km}/N + Cm/(N\ell) = o(s_N).$$

All comparisons remove the spatial mean, which cancels in the difference of the two fixed-size samples.

In continuum units the Gaussian count term for one color is $N^{-1/2} P_\ell W^0$, since the indicator of a cell has L^2 norm ℓ and the raw count standard deviation is $\sqrt{N}\ell$. Independence of the two colors gives $\sqrt{2/N} P_\ell W^0$ for the difference. Thus the empirical and Gaussian histogram gauges differ by $o_{\mathbb{P}}(s_N)$ after this factor. In particular,

$$\mathcal{F}(P_\ell(\mu_N^+ - \mu_N^-)) = \sqrt{2/N}\, \mathcal{F}(P_\ell W^0) + o_{\mathbb{P}}(s_N). \tag{6.5}$$

The Gaussian upper bound and the upper transfer now yield $M_N = O_{\mathbb{P}}(s_N \rho_N)$.

6.3 Reverse inequality and the normalization

Spread the paths of an endpoint-optimal permutation with the same kernel, reflecting each path and its direction as above. Its maximum horizontal displacement is at most $M_N \sqrt{\log(2N)}$ by its exponential modular, and its maximum vertical displacement is at most M_N. Thus $D = O_{\mathbb{P}}(m\rho_N/\sqrt{N}) = o_{\mathbb{P}}(\ell)$. The occupation error $O(D/\ell + s_N/\ell)$, multiplied by M_N, is $o_{\mathbb{P}}(s_N)$. By the occupation estimate, $\int_0^1 r_t\, dt \leq 1 + o_{\mathbb{P}}(1)$ uniformly. The smoothed path current has divergence $S_\ell(\mu_N^+ - \mu_N^-)$. Pointwise Jensen, with zero filler so that the weights have total mass one, bounds its horizontal Luxemburg norm by $(1 + o_{\mathbb{P}}(1))M_N$. Indeed, for an occupation density bound $1 + \epsilon$ one uses the scale $(1 + \epsilon)M_N$. Integrating the Jensen bound in space counts each path with its original weight $1/N$, since the reflected kernel preserves that mass on the square. The vertical supremum has the same factor by direct averaging. Hence

$$\mathcal{F}(S_\ell(\mu_N^+ - \mu_N^-)) \leq M_N + o_{\mathbb{P}}(s_N).$$

By the histogram comparison, the last inequality also holds for $\mathcal{F}(P_\ell(\mu_N^+ - \mu_N^-))$ with an $O_{\mathbb{P}}(s_N)$ error. The multiplicative error in the upper transfer is $O_{\mathbb{P}}(s_N/\ell)$ times a gauge of order $O_{\mathbb{P}}(s_N\rho_N)$, hence it is $o_{\mathbb{P}}(s_N)$. Together with (6.5) this proves (6.1).

Write $\ell = 2^{-j}$, so $j \log 2 = \frac{1}{2}\log N - 10 \log\log N + O(1)$. The uniform Gaussian Lipschitz estimate for $\mathcal{F}$, obtained by applying the grid cut bound (3.1) and affine interpolation, gives

fluctuations $O_{\mathbb{P}}(1)$ around the expected flux gauge. Since $\log\log j \sim \log\log\log N$, this proves (6.2). Theorem 2 now gives

$$M_N = \left(\frac{1}{\sqrt{2\pi}} + o_{\mathbb{P}}(1) \right) \sqrt{\frac{\log N}{N}} \, (\log\log\log N)^{1/4}.$$

The factor $\sqrt{2/N}$ from the difference of the samples and the factor $\sqrt{\frac{1}{2}\log N}$ from the cutoff combine to $\sqrt{\log N/N}$. Reflection in the diagonal of the square preserves the joint sample law and exchanges the coordinate costs, giving the other endpoint.

References

[1] `https://github.com/grottofrancesco/UltimateMatchingTheorem`.